



COMP7503 Multimedia Technologies

Media Sensor and Reproduction - Part 2

Dr. Bill LUO

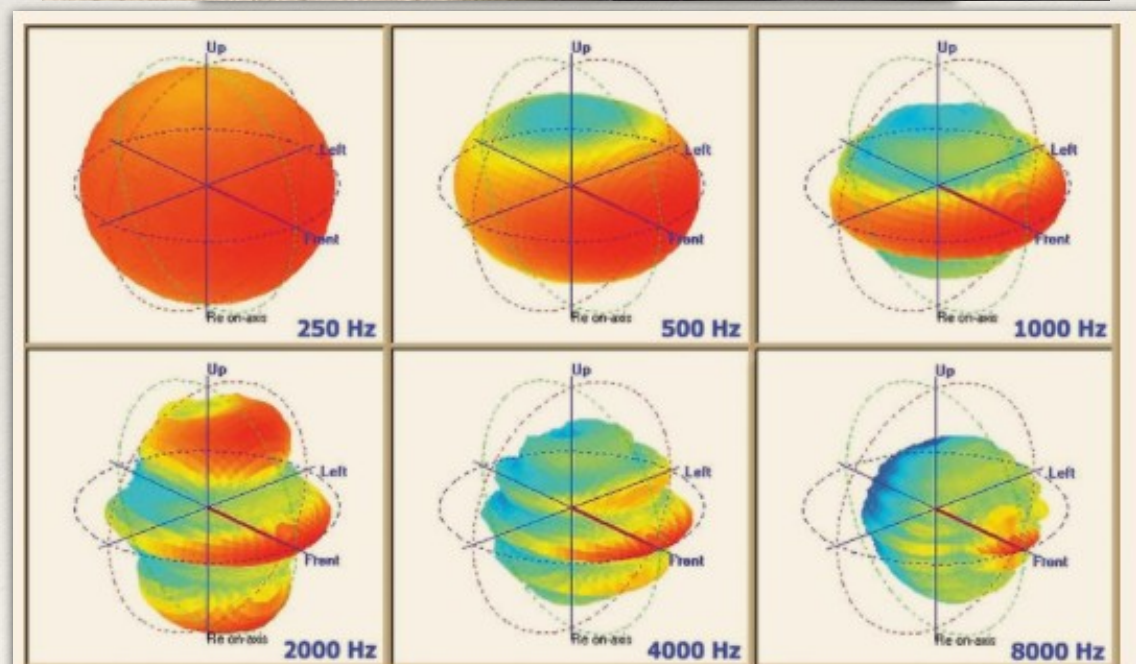


THE UNIVERSITY OF HONG KONG

D E P A R T M E N T O F
COMPUTER SCIENCE

Reproduction of Sound

- ❖ Reproducing sound from storage requires special devices
 - ❖ Loudspeaker (or simply speaker) is the most common sound-reproduction device
- ❖ Loudspeaker performs the reverse operation of a microphone
 - ❖ The speaker pulses or vibrates according to the variations of an electrical signal and causes sound waves to propagate through a medium such as air or water
- ❖ Speakers, like microphones, have directionality as well

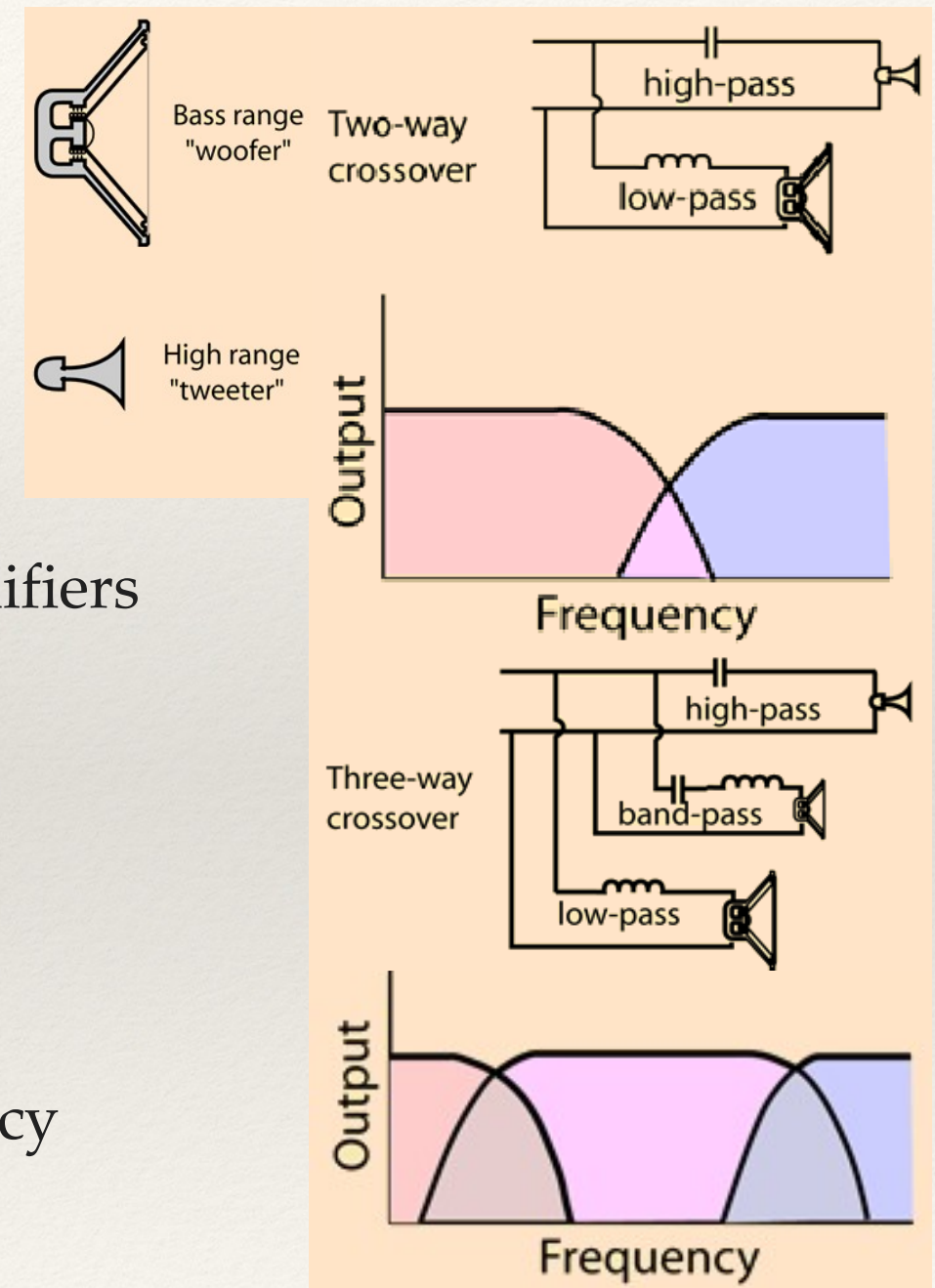


Polar plots of a four-driver industrial columnar public address loudspeaker taken at six frequencies. (Source: <https://en.wikipedia.org/wiki/Loudspeaker>)



Speaker Characteristics

- ❖ Rated power
 - ❖ Maximum input a speaker can take before it is destroyed
- ❖ Impedance
 - ❖ The electrical compatibility with different amplifiers
- ❖ Crossover frequencies
 - ❖ Signal division by drivers
- ❖ Frequency range
 - ❖ Determines the speaker system's linear frequency response range



Reference: <http://hyperphysics.phy-astr.gsu.edu/hbase/audio/cross.html>



Crossover Frequencies

- ❖ Different sound frequencies should be better driven by different sized speakers
 - ❖ For low frequencies (base) it is easier to get something with a bigger diaphragm to vibrate slower because of its larger mass, and vice versa

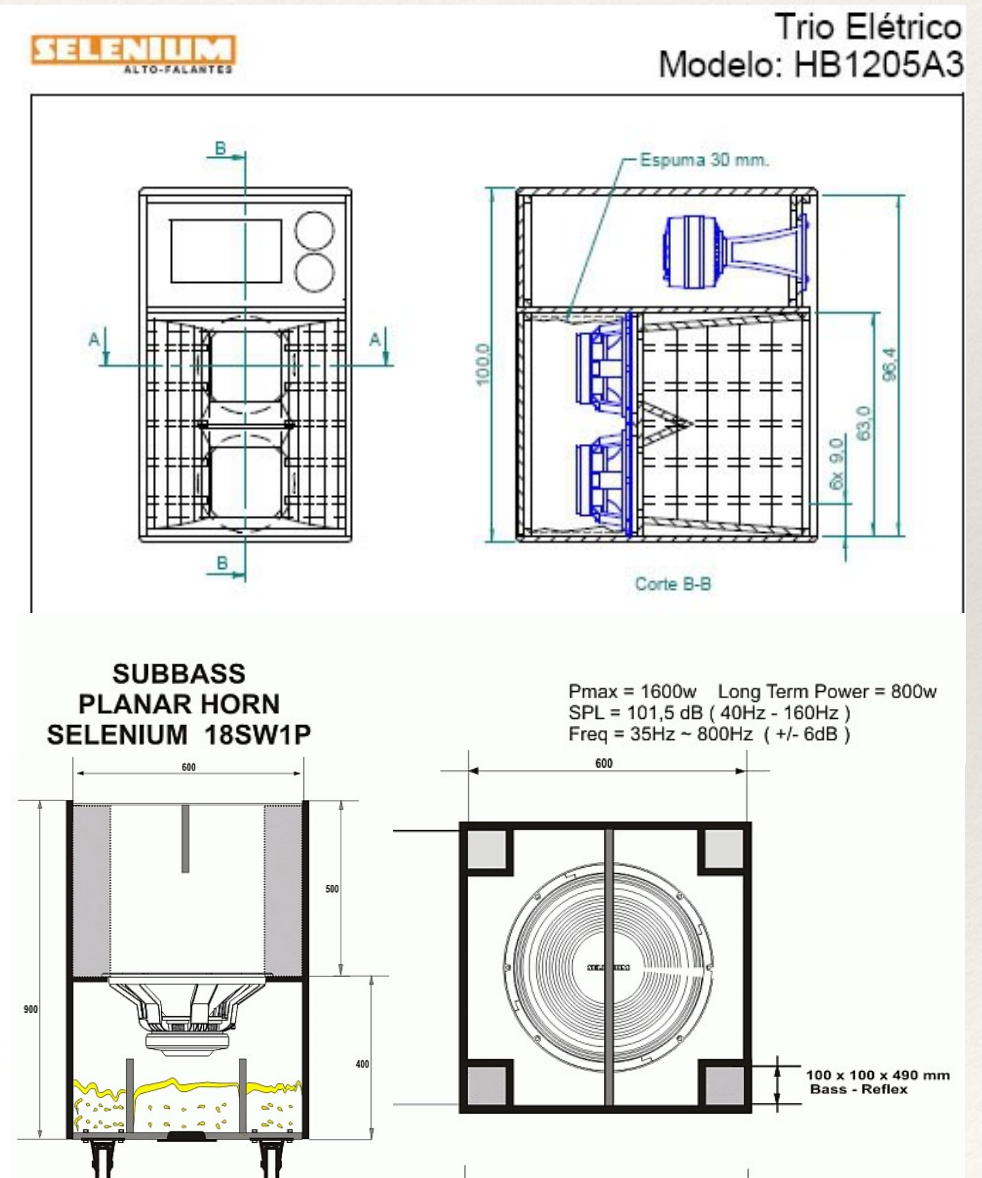
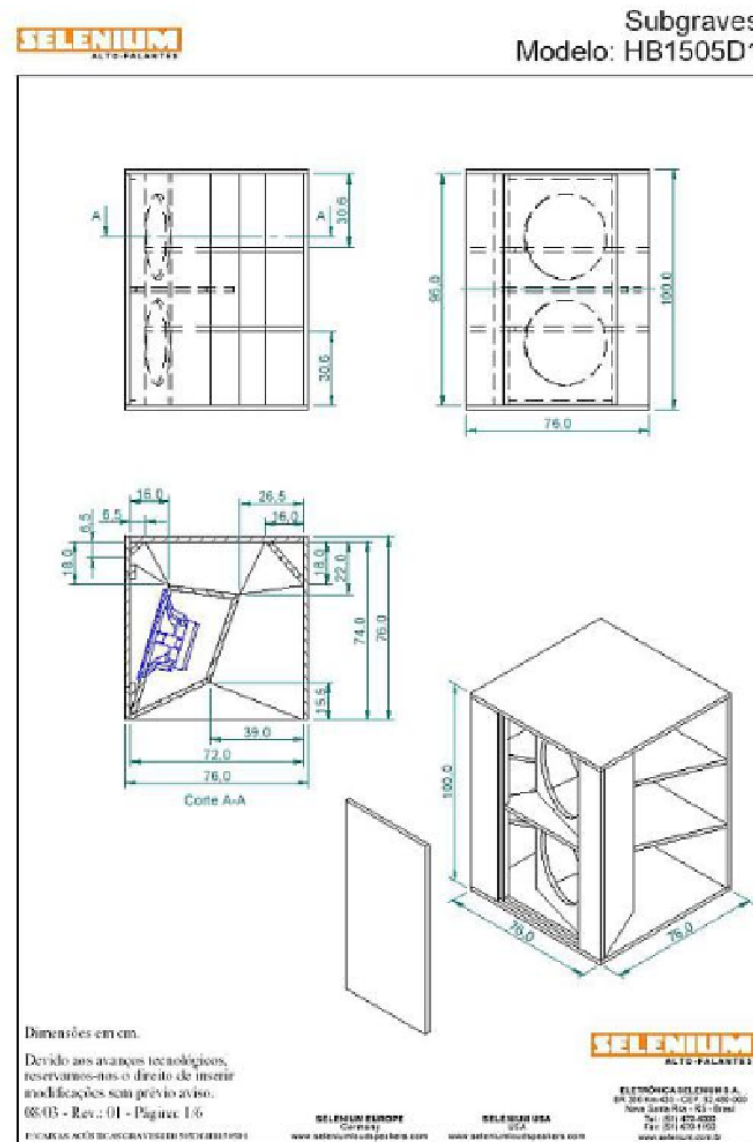
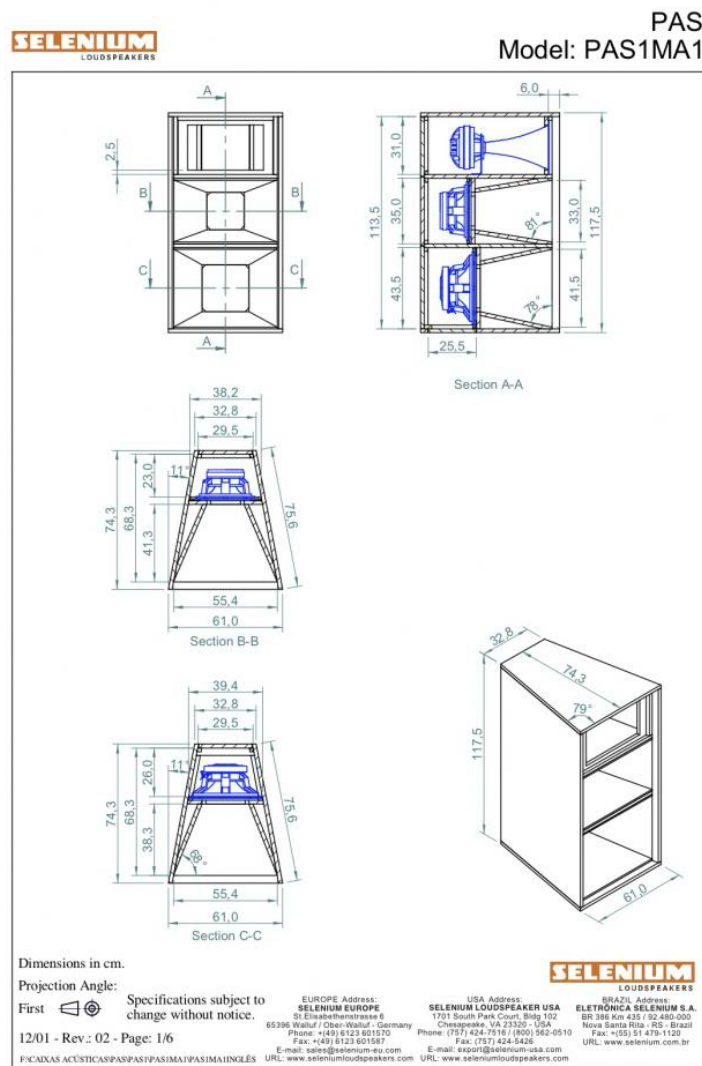


Reference: <http://www.sony.com/electronics/wireless-speakers/srs-x88>



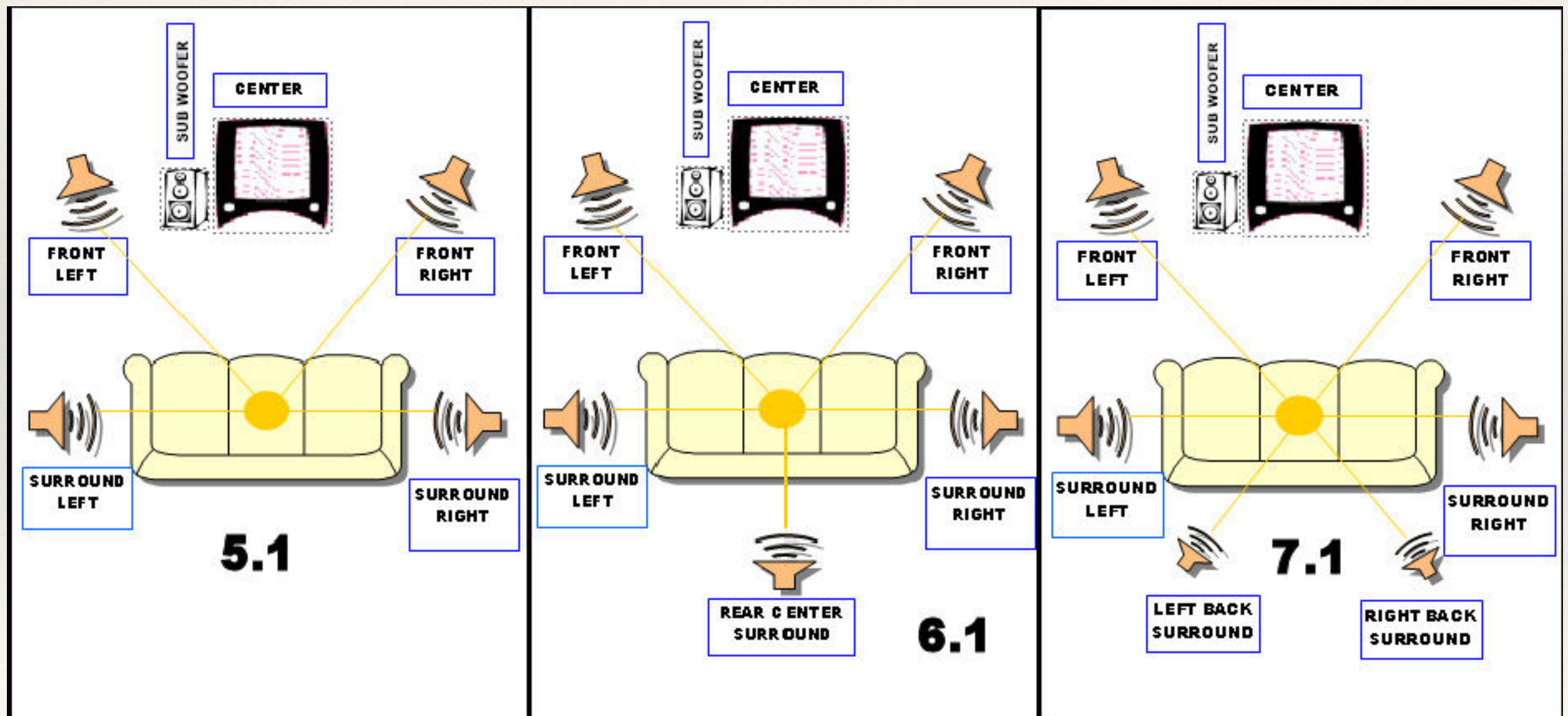
Speaker Enclosures

- ❖ The primary role of the enclosure is to prevent sound waves generated by the rearward-facing surface of the diaphragm interacting with sound waves generated at the front of the driver



Surround Sound

- ❖ Surround Sound is commonly used to produce sound output that appears to surround the listener — that is, to provide a more realistic and engaging experience



Reference: <http://www.webopedia.com/DidYouKnow/Computer Science/surround sound.asp>

Virtual Surround Sound

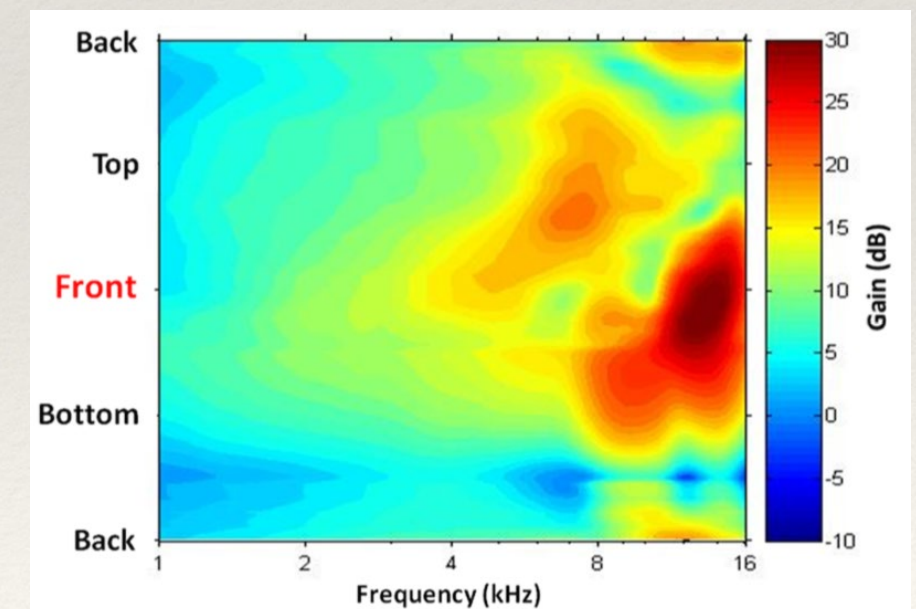
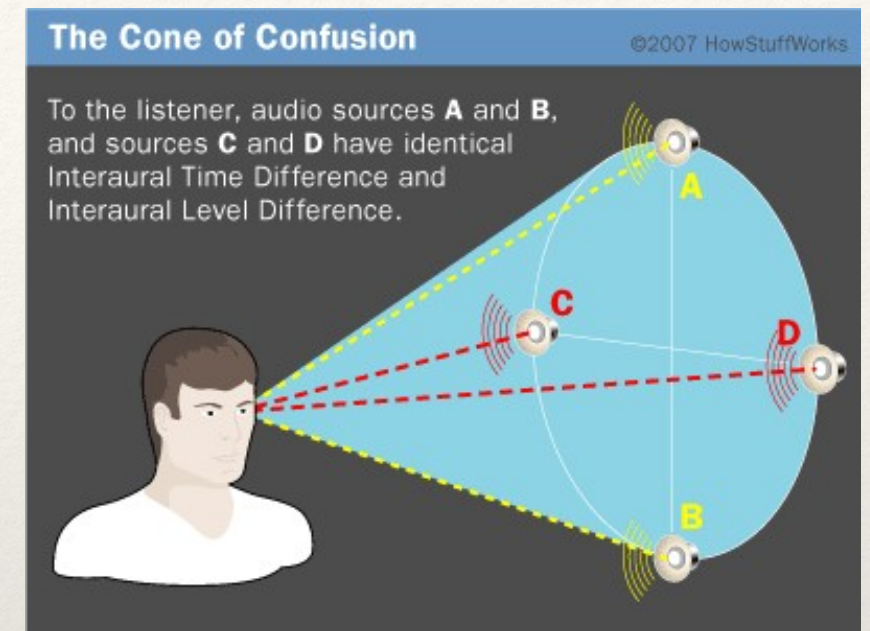
- ❖ 5.1, 6.1 and 7.1 are sometime difficult to deploy due to space and wiring constraints
- ❖ Virtual Surround Sound mimics the effect of a multispeaker surround-sound system
 - ❖ Two general configurations
 - ❖ 2.1 surround
 - ❖ Simulate 5.1 based on two front speakers and one subwoofer
 - ❖ Digital sound projection 
 - ❖ Simulate 5.1 based on a single strip of small speakers

Reference: <http://electronics.howstuffworks.com/virtual-surround-sound3.htm>



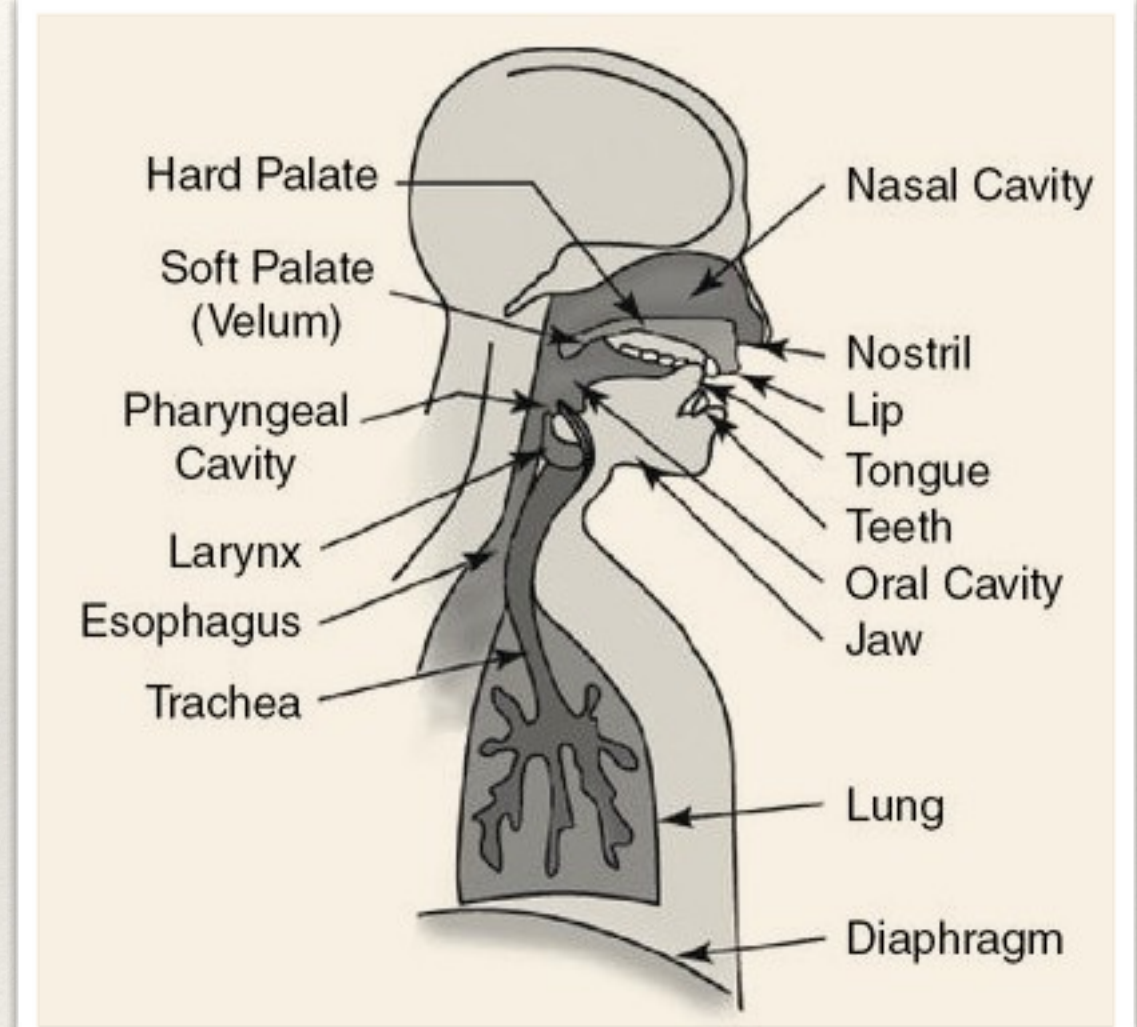
How to produce Virtual Surround Sound?

- ❖ Between left and right ears
 - ❖ The difference in audio volume is the Interaural Level Difference (ILD)
 - ❖ The difference in arrival time (delay) is the Interaural Time Difference (ITD)
- ❖ ILD and ITD can be analysed by our brain to determine whether the sound came from the left or right
- ❖ In addition, human brain can use a sound wave's reflection off the pinna, or auricle, or the ear to determine the sound's location
 - ❖ The volume and sound quality change due to these reflections are known as Head-Related Transfer Functions (HRTFs)
- ❖ To produce virtual surround sound
 - ❖ Recreate the ILD, ITD, HRTFs based on the speakers available



Speech Production

- ❖ Frequency range of normal human speech is 80 Hz ~ 5 kHz
- ❖ The pitch of human voice
 - ❖ Adult Male: 120 ~ 160 Hz
 - ❖ Women and Children: 220 ~ 330 Hz
- ❖ Vowels can reach frequencies up to 5 KHz
- ❖ Sibilants can emit frequencies as high as 20 kHz
- ❖ Volume of human voice at 60cm from the mouth
 - ❖ 60 dBA for normal speech, yelling at 76 dBA for males and 68 dBA for females



Source: National Cancer Institute

Speech Production (Cont'd)

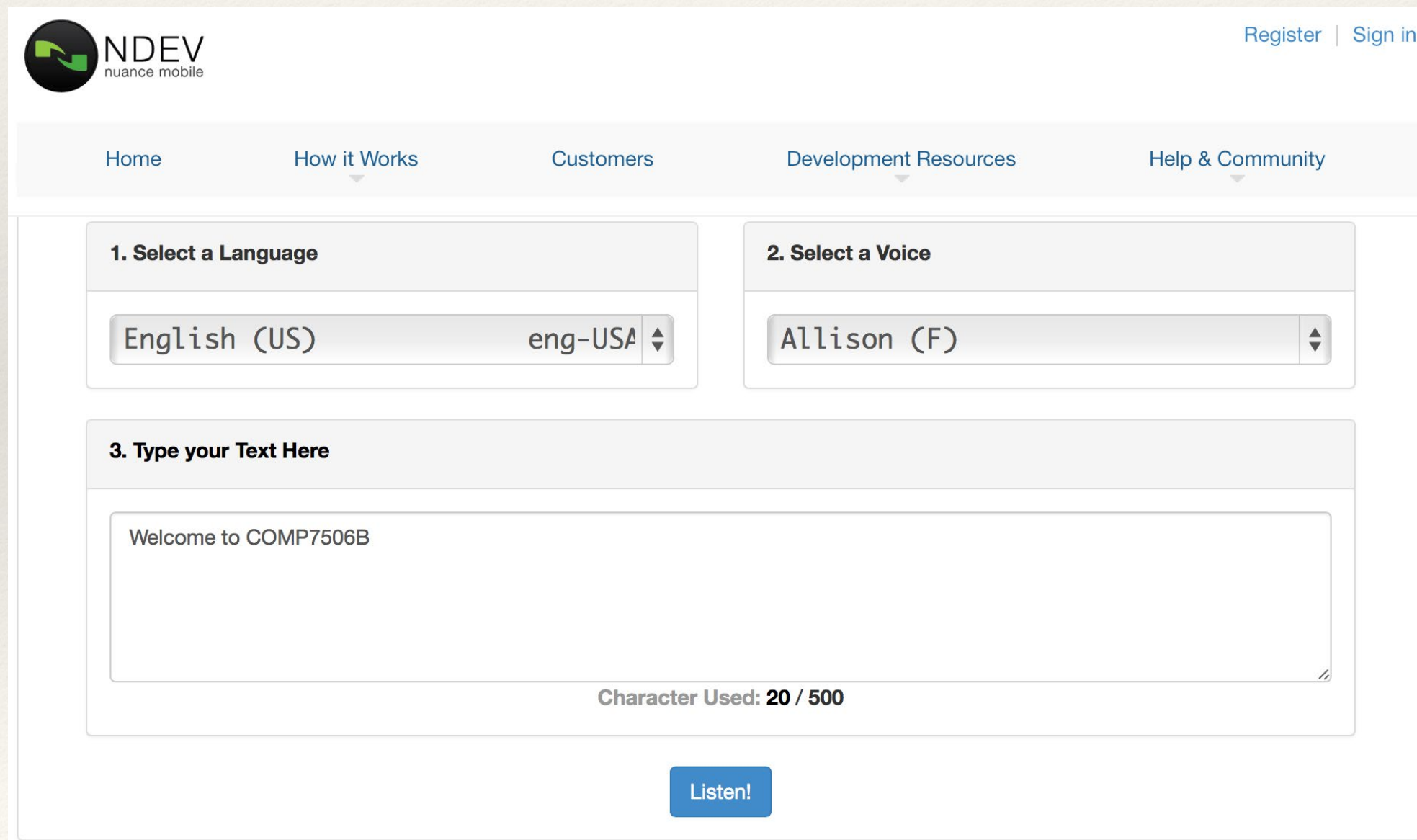
- ❖ Speech sounds are usually classified as
 - ❖ Stop consonants
 - ❖ With blocked airflow, such as the English pronunciations of “p,” “t,” or “k”,
 - ❖ Fricative consonants
 - ❖ With partially blocked and therefore strongly turbulent airflow, such as the English “f” or “v”
 - ❖ Approximants
 - ❖ With only slight turbulence, such as the English “w” and “r”
 - ❖ Aspirates
 - ❖ Pronounce with an exhalation of breath, the sound of *h* at the beginning of a word
 - ❖ Vowels
 - ❖ With full, unimpeded airflow, such as the English “a,” “e,” “i,” and “o”)
- ❖ Consonants and vowels are the building blocks of speech, usually referred to as *phonemes*
- ❖ Variations exists for different languages



Speech Synthesizer

- ❖ Nuance Mobile Text-to-Speech Engine

- ❖ <http://dragonmobile.nuancemobiledeveloper.com/public/index.php?task=supportedLanguages>



The screenshot shows the NDEV Nuance Mobile Text-to-Speech Engine interface. At the top left is the NDEV logo with the text "nuance mobile". At the top right are links for "Register" and "Sign in". Below these are navigation tabs: "Home", "How it Works", "Customers", "Development Resources", and "Help & Community". The main content area is divided into three sections: 1. Select a Language, 2. Select a Voice, and 3. Type your Text Here. In the first section, "English (US)" is selected with a dropdown arrow showing "eng-USA". In the second section, "Allison (F)" is selected with a dropdown arrow. The third section contains a text input field with the text "Welcome to COMP7506B" and a character count "Character Used: 20 / 500". At the bottom center is a blue "Listen!" button.

NDEV
nuance mobile

Register | Sign in

Home How it Works Customers Development Resources Help & Community

1. Select a Language

English (US) eng-USA

2. Select a Voice

Allison (F)

3. Type your Text Here

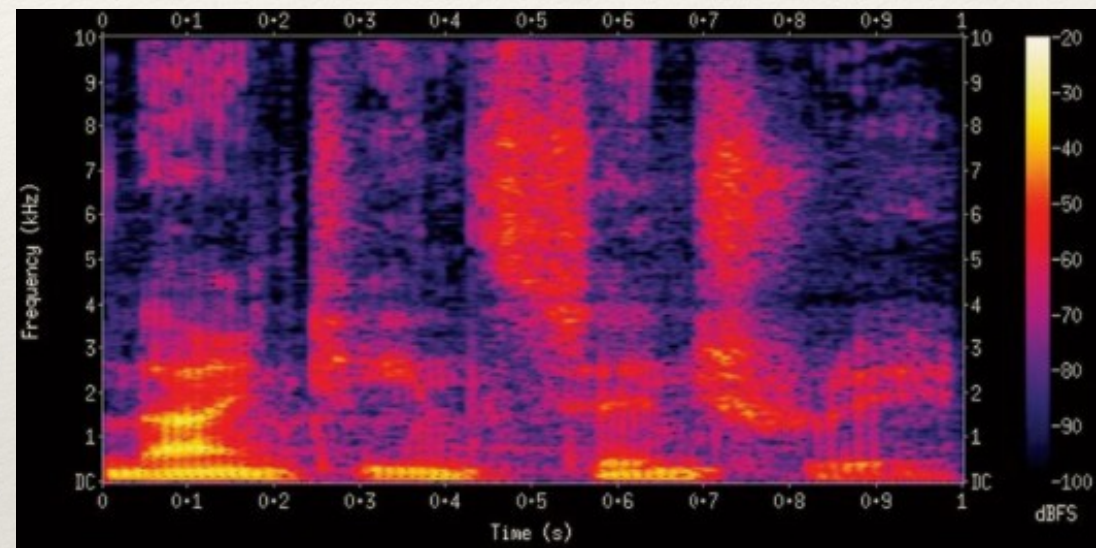
Welcome to COMP7506B

Character Used: 20 / 500

Listen!

Speech Recognition

- ❖ Each phoneme is distinguished by its own unique pattern in the spectrogram
- ❖ For voiced phonemes, the signature involves large concentration of energy, called formants
 - ❖ See yellow bands in the spectrogram



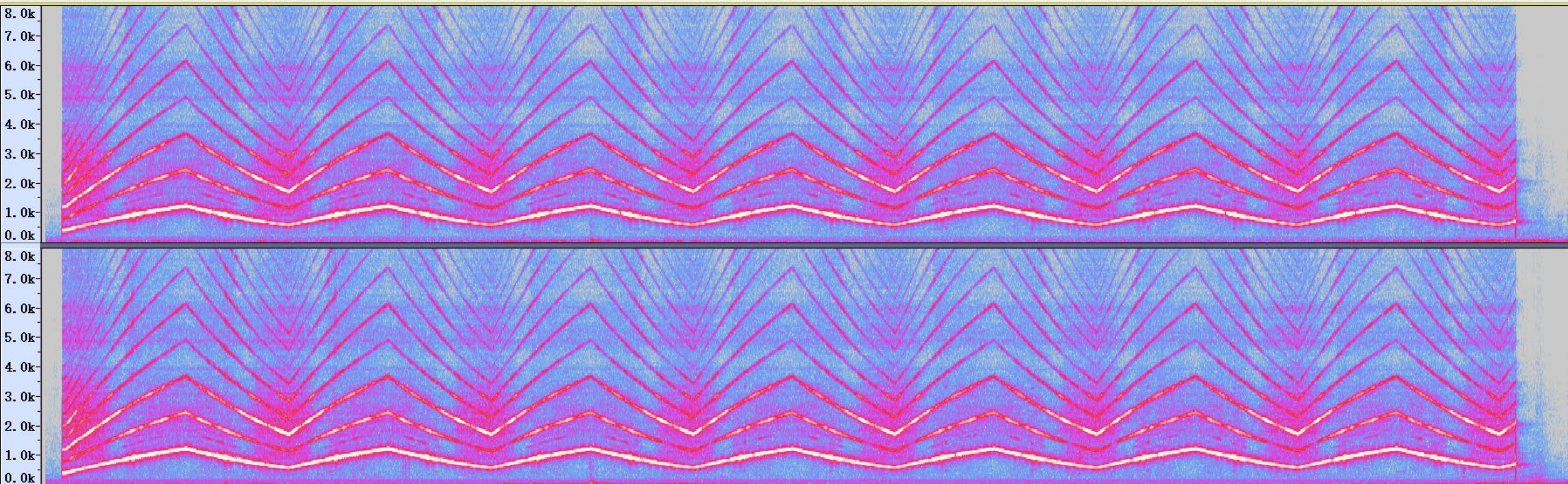
Source: Wikimedia Commons

- ❖ Unvoiced speech does not reflect in the spectrogram as formants
 - ❖ Stop consonants can be detected as strong bursts of energy across all frequency bands occurring after a short silence
 - ❖ Fricatives, Aspirates can be recognized as “large clouds” of smooth energy



Spectrogram

<https://academo.org/demos/spectrum-analyzer/>

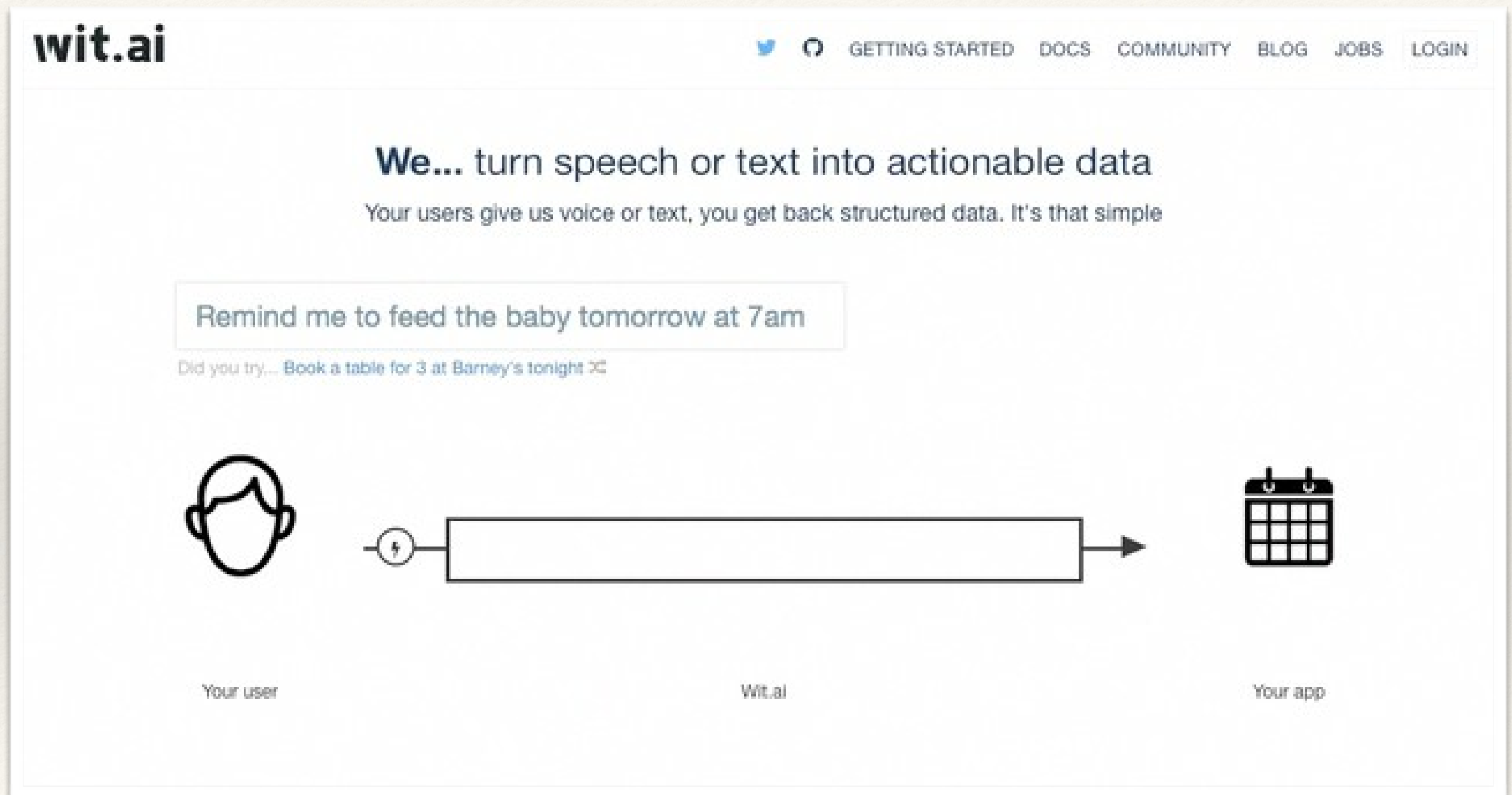


Speech Recognition Limitations

- ❖ Human tend to increase volume, change pitch, or adjust duration of sound of syllables in response to external noise
 - ❖ Automatic speech recognition algorithms trained in a quiet environment are difficult to transform to noisy environments
- ❖ The highest frequencies are emitted by sibilants, such as “s” or “f”, can be well above 20 kHz
 - ❖ Conversational speech were usually sampled at 8 KHz or below, so it is usually difficult to differentiate these sibilants on phone
- ❖ Solution
 - ❖ Spelling performed in whole words: “Alpha, “Beta”, “Charlie”, ... “Frankie”, ... , “Sugar”, ..., “Zero”

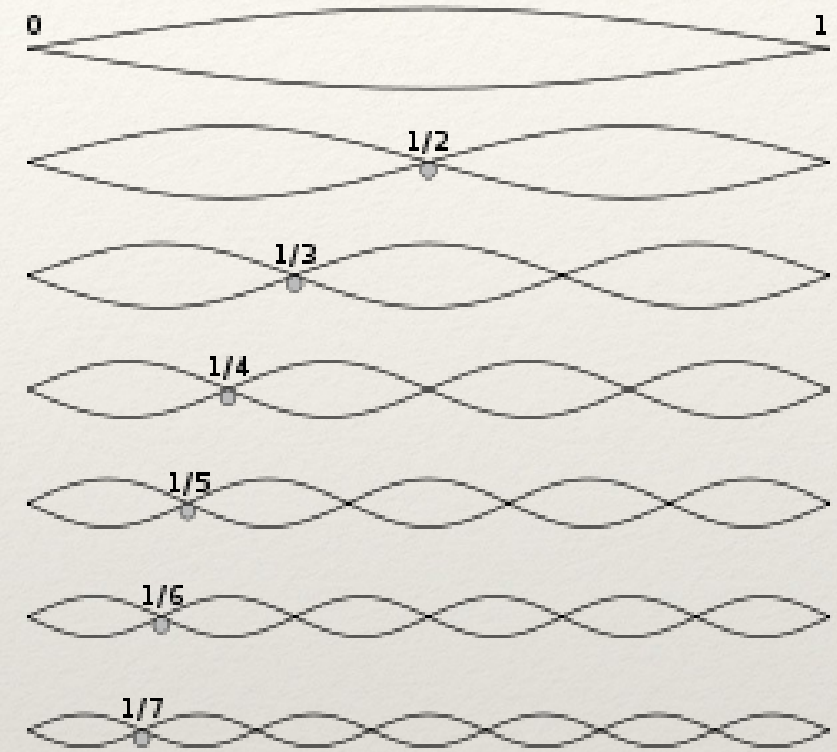


Dictation + Natural Language Processing + Actionable Intelligence



Music

- ❖ In multimedia computing, we are interested in determining musical instruments' general properties, so that we can
 - ❖ Compress audio
 - ❖ Detect Instruments
 - ❖ Manipulate or Synthesise Music
- ❖ Pitch represents a sound's perceived fundamental frequency
 - ❖ Fundamental frequency is denoted as f_0 or F_0
- ❖ Overtone
 - ❖ A harmonic or partial resonance



Frequency	Order	Name 1	Name 2	Name 3
$1 \cdot f = 440 \text{ Hz}$	$n = 1$	fundamental tone	1st harmonic	1st partial
$2 \cdot f = 880 \text{ Hz}$	$n = 2$	1st overtone	2nd harmonic	2nd partial
$3 \cdot f = 1320 \text{ Hz}$	$n = 3$	2nd overtone	3rd harmonic	3rd partial
$4 \cdot f = 1760 \text{ Hz}$	$n = 4$	3rd overtone	4th harmonic	4th partial

Reference: <https://en.wikipedia.org/wiki/Overtone>



Music (Cont'd)

❖ Timbre

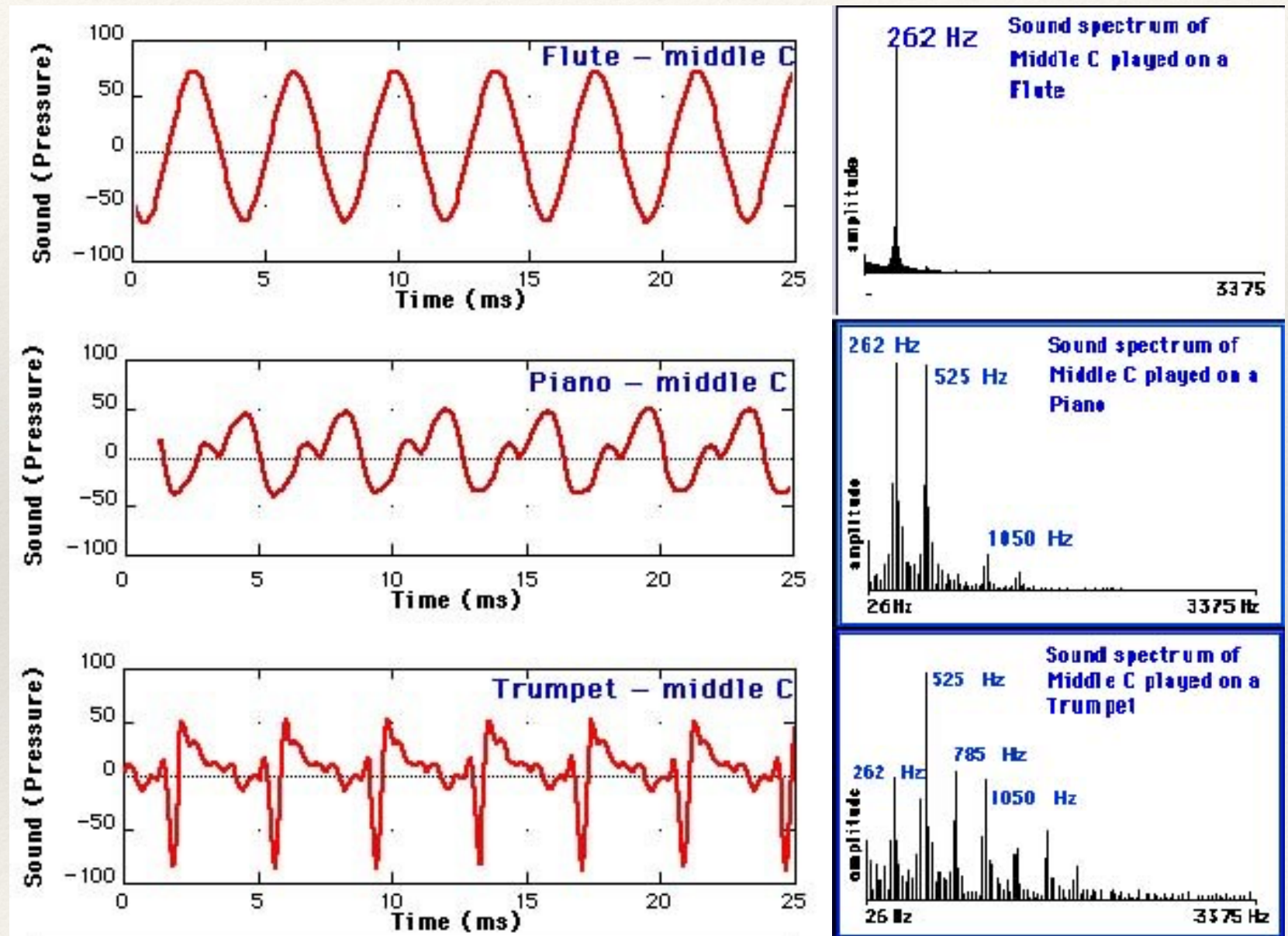
- ❖ It describes the quality of sound that distinguishes different types of sound production
- ❖ Characterised by frequency spectrum and time envelope

❖ Spectrum

- ❖ The sum of distinct frequencies emitted by an instrument playing a particular note

❖ ADSR envelope

- ❖ Four stages of sound
 - ❖ Attack, Decay, Sustain, and Release

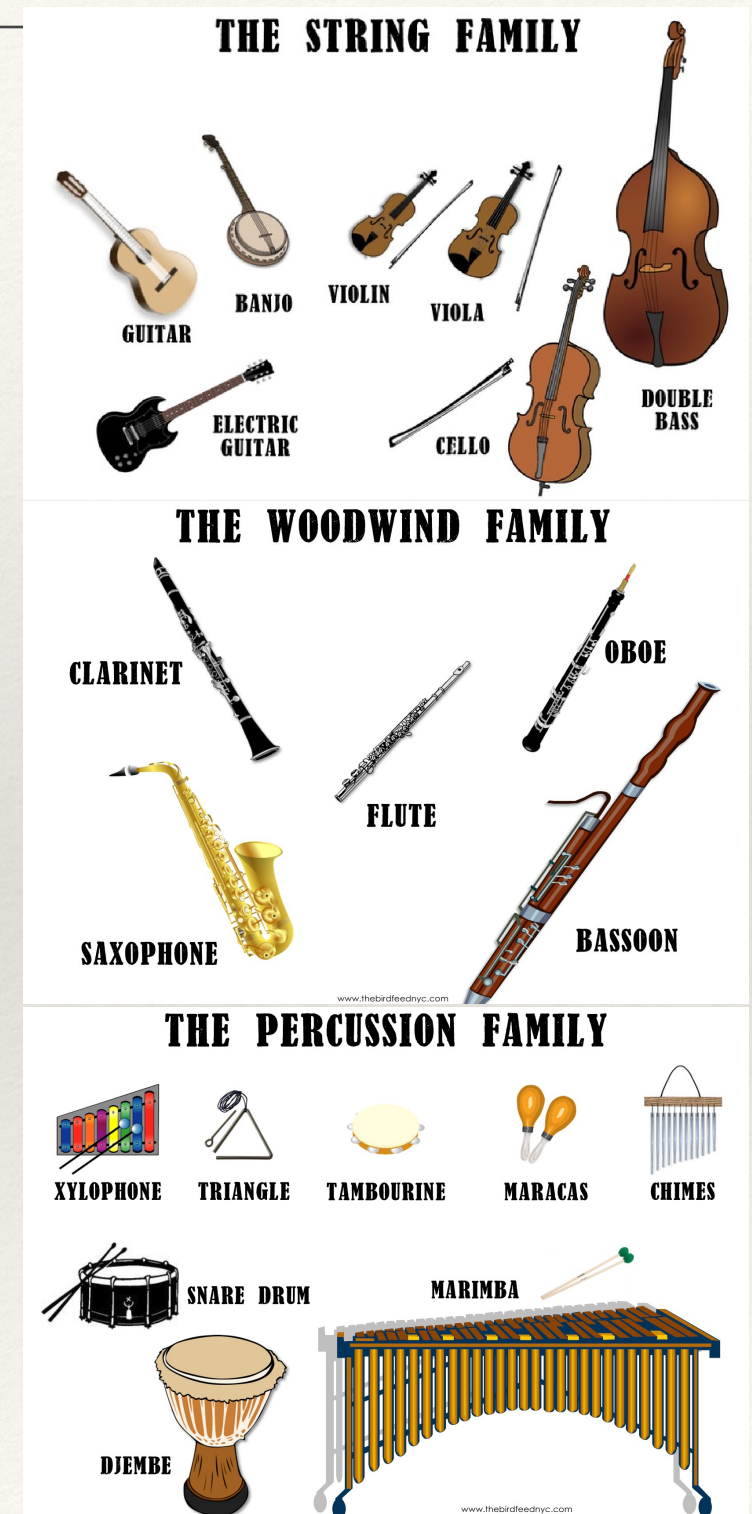


Source: http://castle.eiu.edu/~ddavis/chapter_12/ch12_3.htm



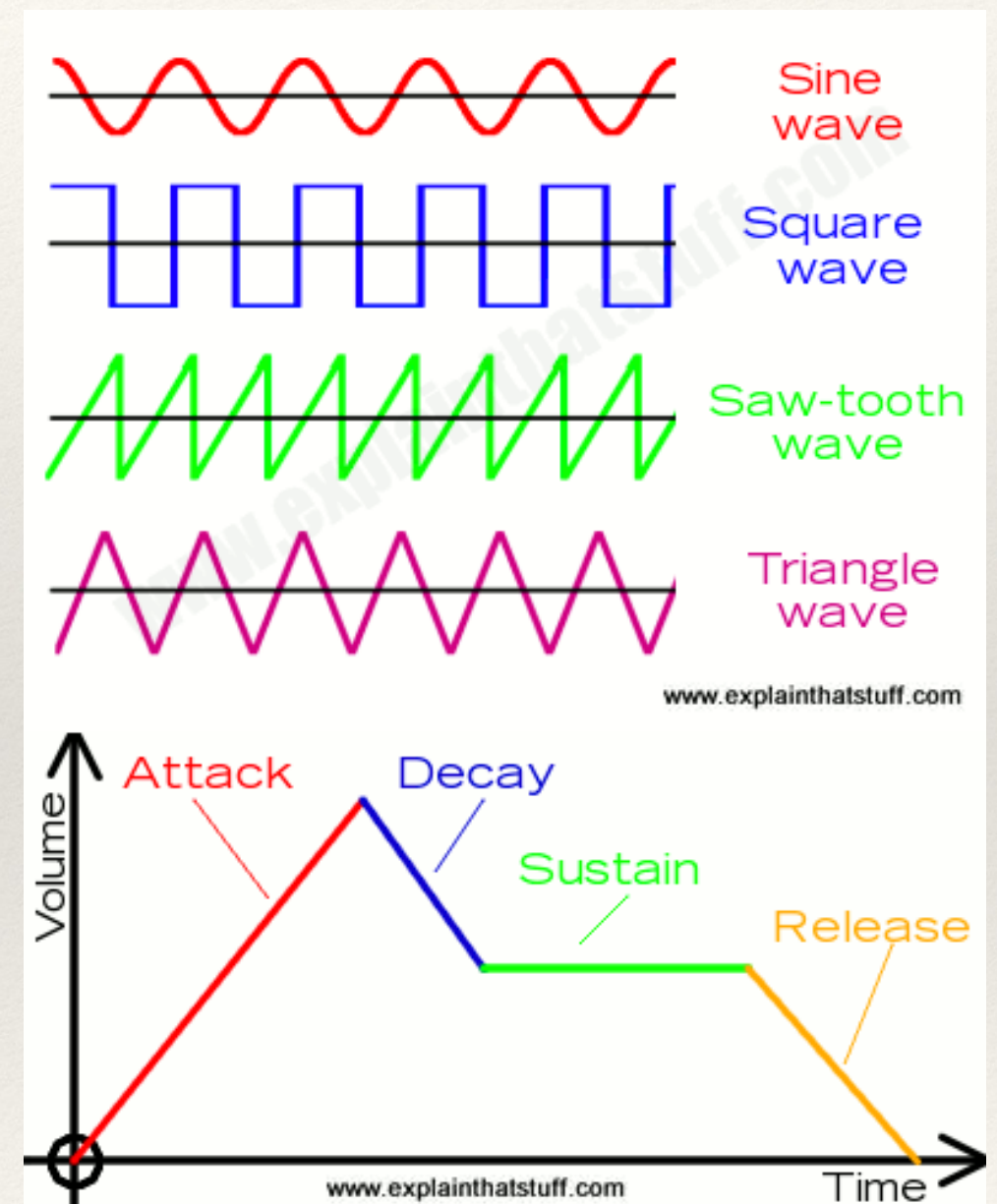
Categories of Musical Instruments

- ❖ String - Produce sound by vibrating strings
 - ❖ Particular mix of harmonics present depends on the method of excitation
- ❖ Wind - Produce sound by blowing into the end of the resonator
- ❖ Percussion - Produce sound by hitting, shaken, rubbing, scraped



Music Synthesizer

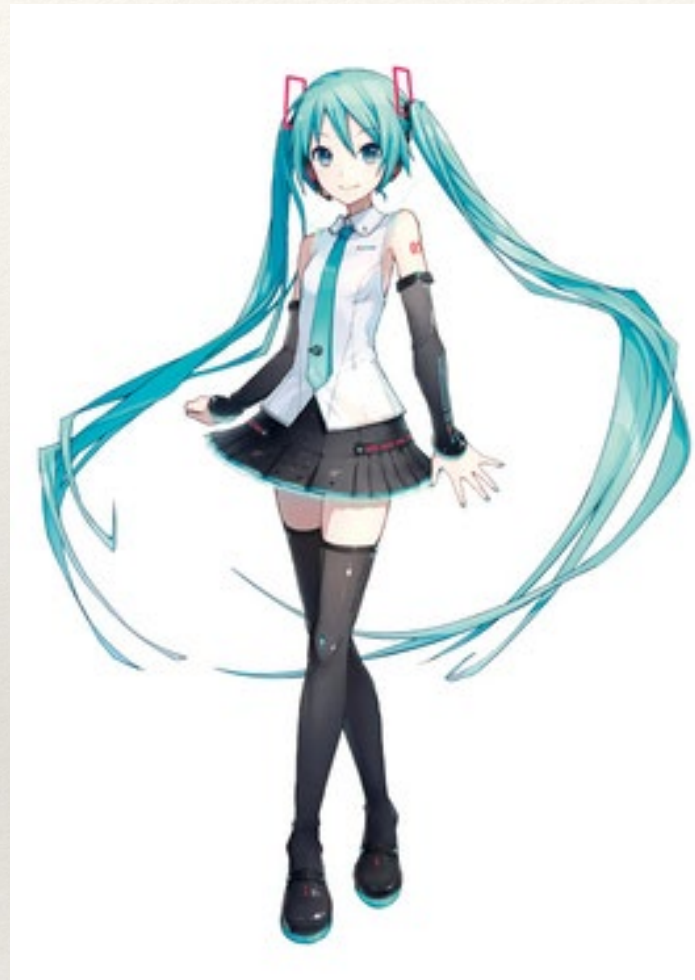
- ❖ Music synthesis is performed by
 - ❖ Physical modelling of the instrument
 - ❖ Harmonics
 - ❖ Wave envelope
 - ❖ i.e. ADSR
 - ❖ Incorporating original samples of the instrument
 - ❖ That is, the wavetables
- ❖ Basic Principle
 - ❖ Generate sound waves of different shapes.
 - ❖ Generate more than one sound tone at once to produce a fundamental frequency and harmonics.
 - ❖ Make the volume of the sound change over time to produce different ADSR envelope shapes.



Reference: <http://www.explainthatstuff.com/synthesizers.html>



Music Synthesizer



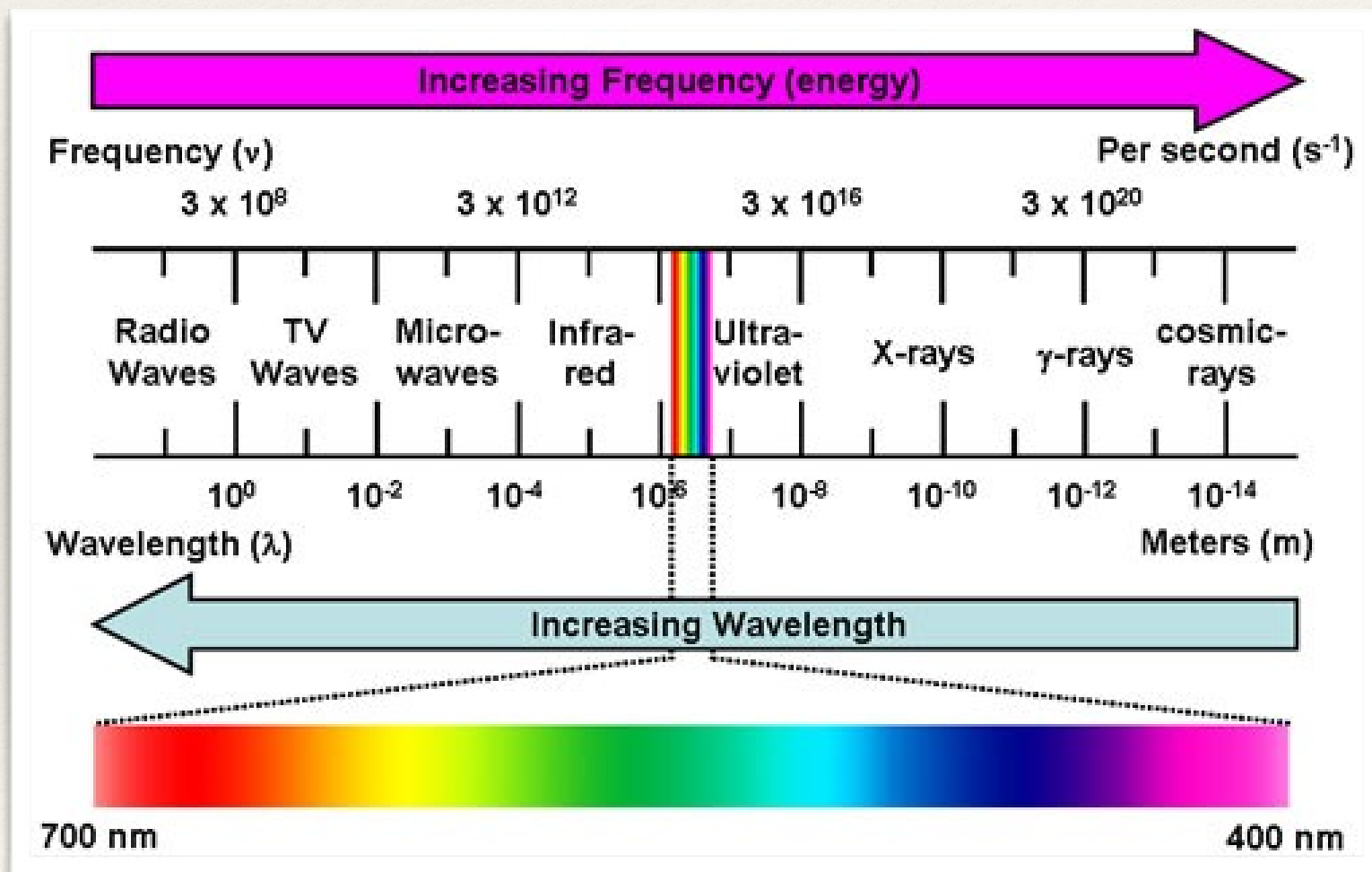
Hatsune Miku

Ref: https://en.wikipedia.org/wiki/Hatsune_Miku

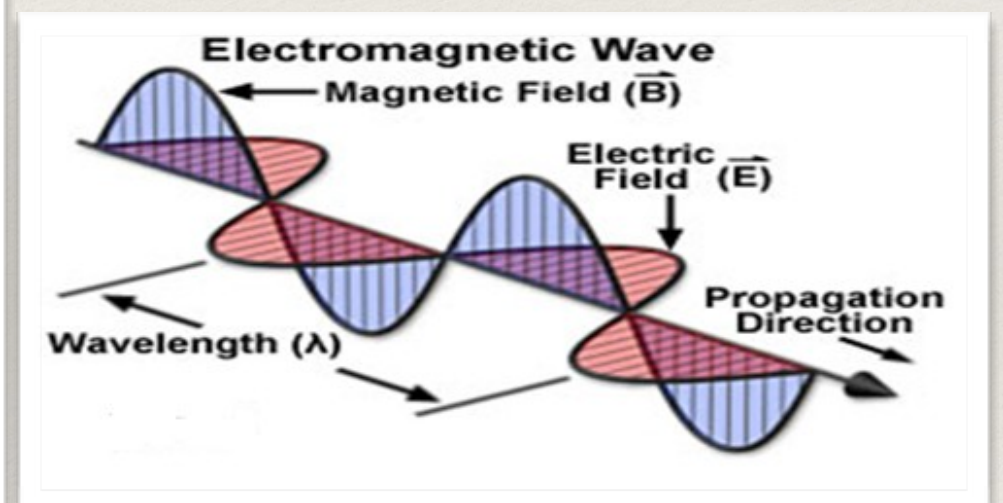


Light

- ❖ It is an electromagnetic wave which has wavelength range from 400 to 780 nm



Source: <http://www.chemicalconnection.org.uk/chemistry/topics/images/ee5c.jpg>



Source: <https://rooksheathscience.files.wordpress.com/2013/03/image2.png>

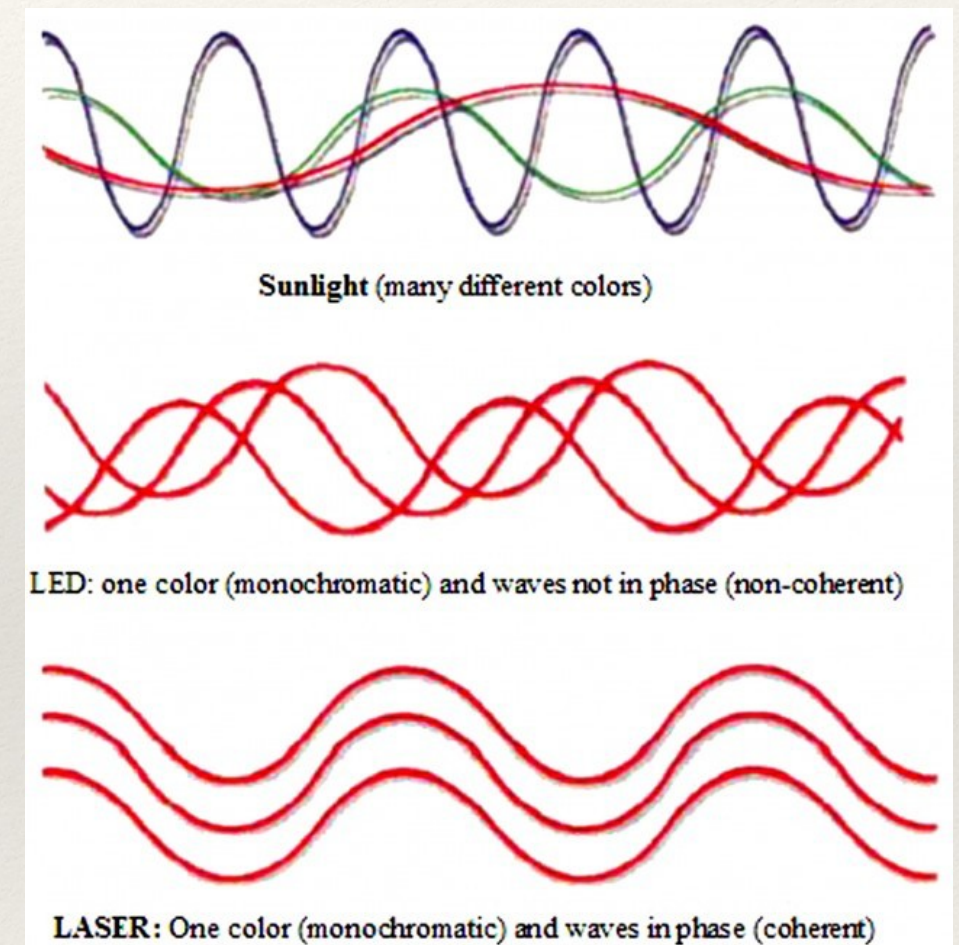
Light (Cont'd)

❖ Speed of Light (c)

- ❖ 299,792,458 meters per second in vacuum
- ❖ Usually approximated as $c = 3.00 \times 10^8 \text{ ms}^{-1}$

❖ Phase

- ❖ Light from normal light source are usually not in phase
 - ❖ i.e. Non-coherent



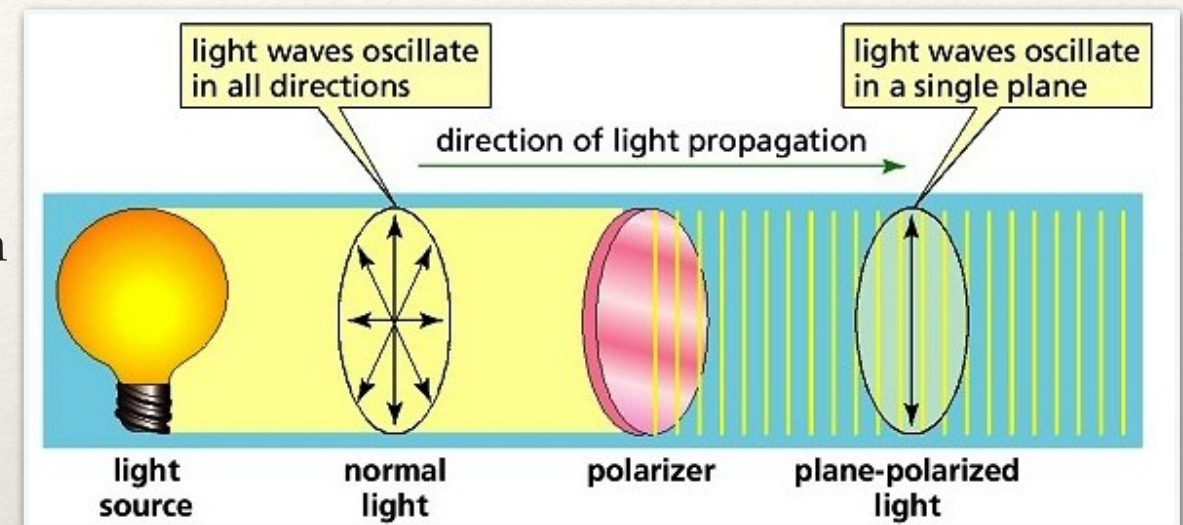
Source: <http://thescienceofacne.com/wp-content/uploads/multichromatic-sunlight-monochromatic-and-coherent-light-640x652.jpg?3225c4>



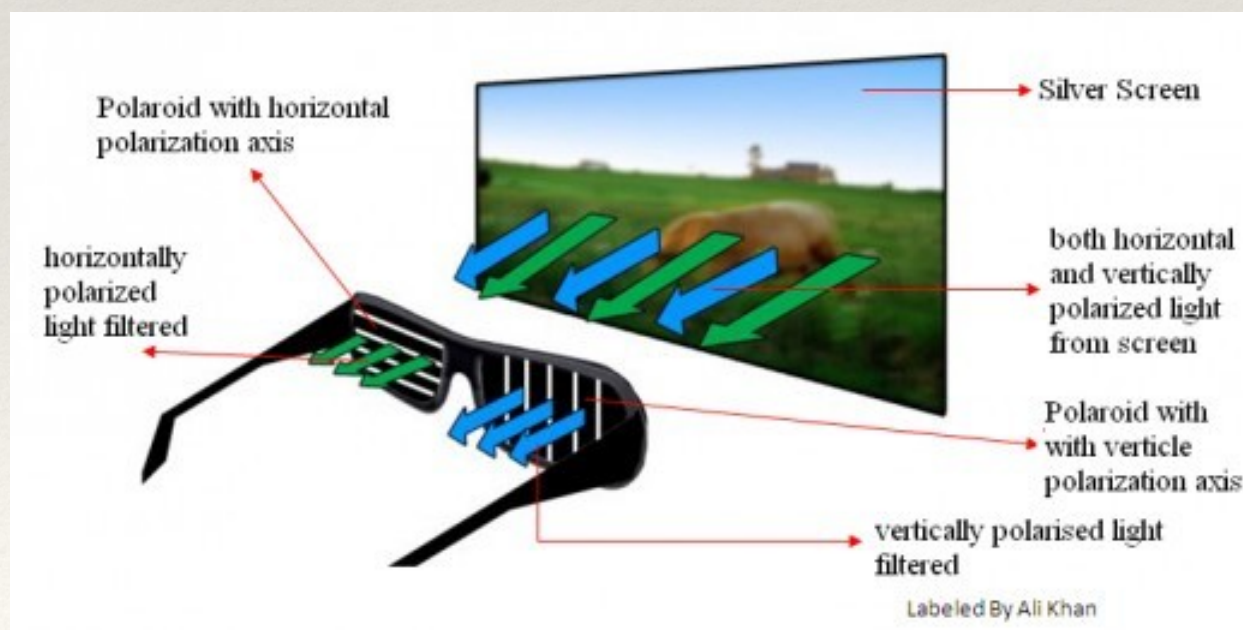
Light (Cont'd)

❖ Polarization

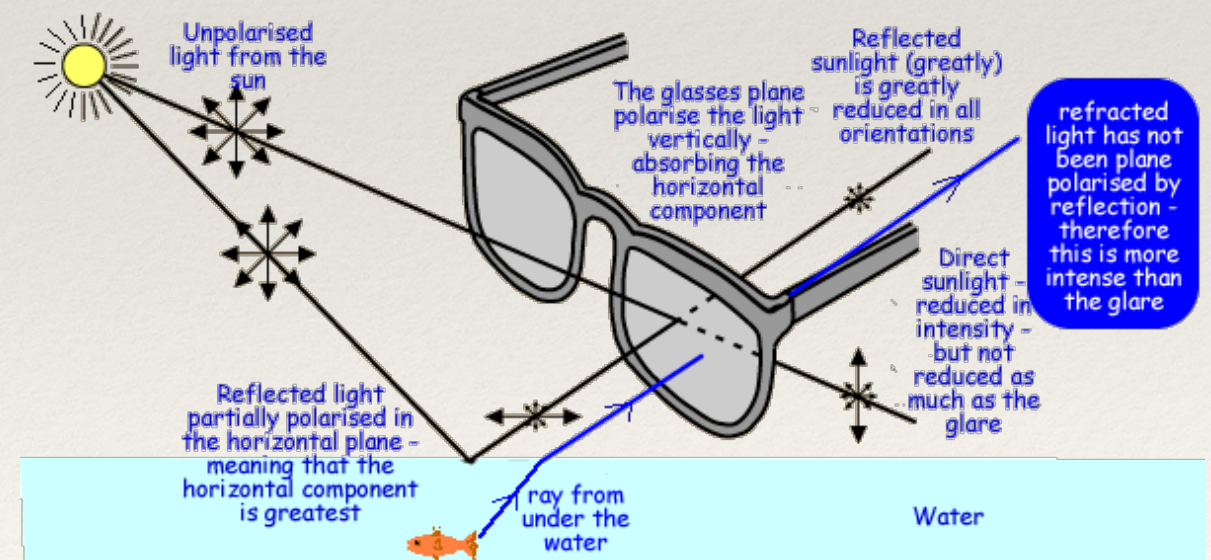
- ❖ It describes the light waves' orientation
- ❖ If the orientation of the electric fields emitted from the light source are not correlated, the light is said to be unpolarized
- ❖ Filters can be used to allow light of a certain degree and angle of polarisation to pass through



Source: <http://www.kshitij-iitjee.com/Study/Chemistry/Part2/Chapter3/36.jpg>



Source: http://usercontent1.hubimg.com/6244450_f520.jpg



Source: <http://www.cyberphysics.co.uk/graphics/diagrams/light/sunglasses.gif>

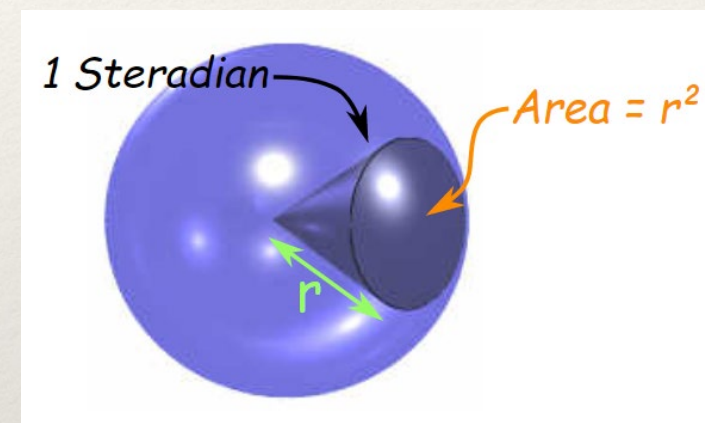
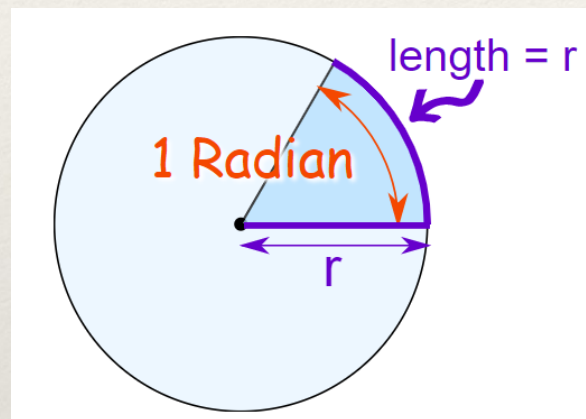
Light (Cont'd)

- ❖ Light intensity is measured in three units:
 - ❖ Candela (*cd*)
 - ❖ Measures luminous intensity, which is defined as power emitted by a light source in a particular direction
 - ❖ Lumen (*lm*)
 - ❖ It measures luminous flux, i.e. the total light an object emits
 - ❖ Lux (*lx*)
 - ❖ It measures the luminous power per area



Candela (cd)

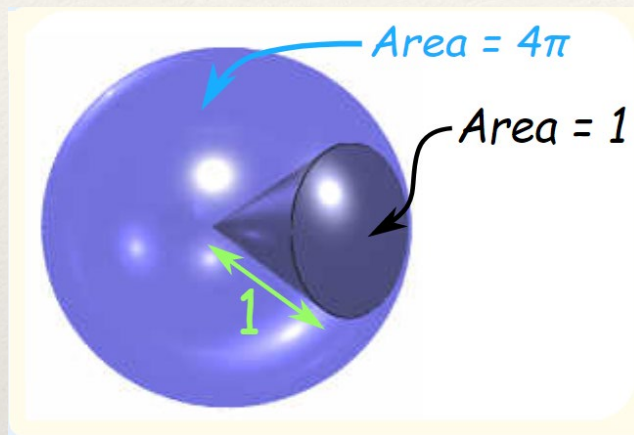
Radiant intensity? Radiant?



A Radian "cuts out" a length of a circle's circumference equal to the radius.

A Steradian "cuts out" an area of a sphere equal to $radius^2$.

Candela (cd)



Example: You measure the light coming from a powerful globe.

Your sensor is 50mm × 50mm in size, and if you hold it 2m away it measures 0.1 Watts.

What is the radiant intensity in **W/sr** (Watts per steradian)?

Answer: At 2m, one steradian cuts through $2 \times 2 = 4 \text{ m}^2$ of the sphere.

And because the sensor is relatively small, its flat surface area is approximately the area of sphere that it occupies. So $0.05 \times 0.05 = 0.0025 \text{ m}^2$.

So, one steradian receives about $0.1 \text{ W} \times (4 \text{ m}^2 / 0.0025 \text{ m}^2) = \mathbf{160 \text{ W/sr}}$.

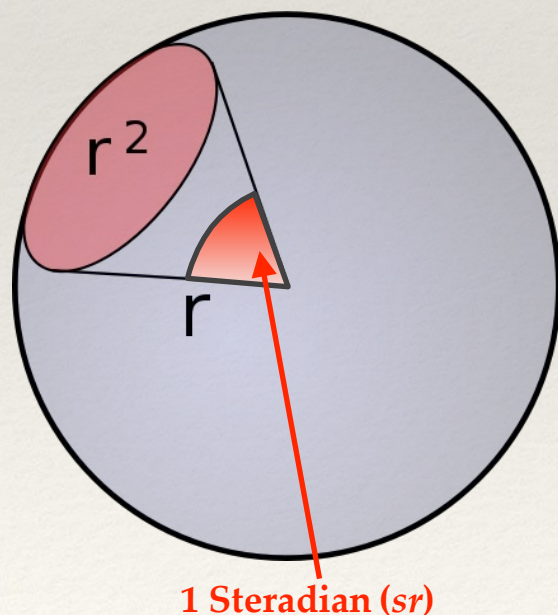
The "unit sphere":

- has a radius of $r(1)$
- has a surface area of $4\pi r^2$ (4π),
- a steradian "cuts out" an area equal to $radius^2$ (1).



Candela (cd)

- ❖ One candela is the luminous intensity, in a given direction and weighted by the luminosity function, of a source that emits monochromatic radiation of frequency
 - ❖ 540×10^{12} hertz, and
 - ❖ Has a radiant intensity in that direction of $1/683$ watt per steradian (sr)
- ❖ A 100-watt incandescent light bulb emits about 120 cd

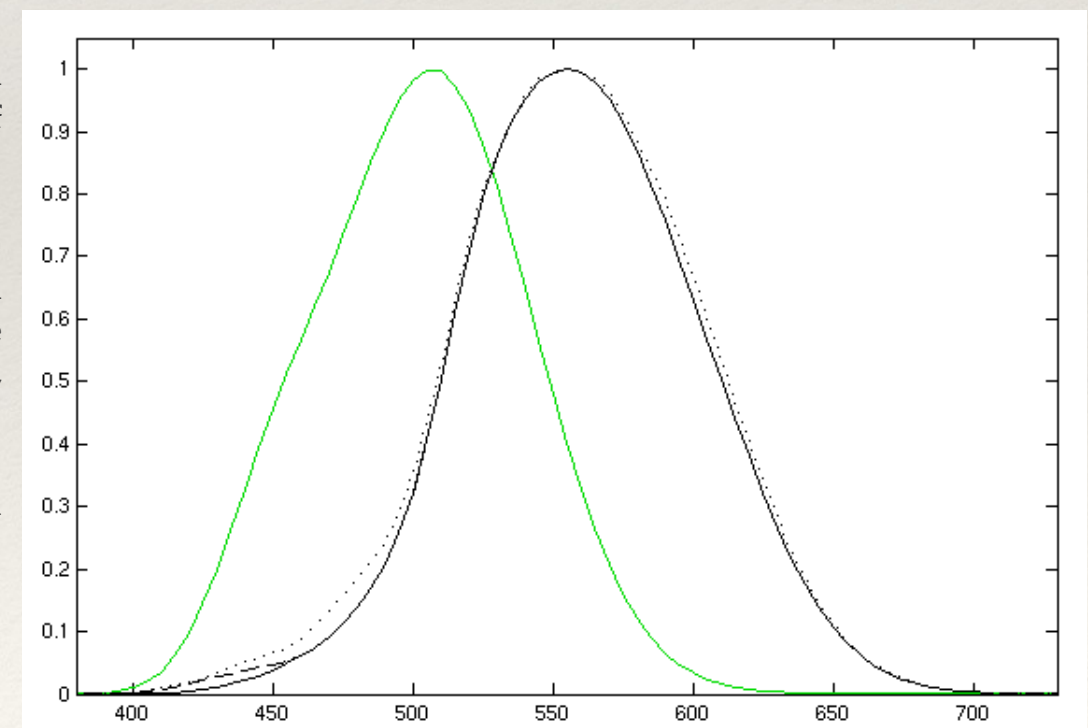


The luminosity function or luminous efficiency function describes the average spectral sensitivity of human visual perception of brightness

Graph of photopic luminosity function (black) including CIE 1931 (solid), Judd-Vos modified (dashed), and Sharpe, Stockman, Jagla & Jägle 2005 (dotted); and scotopic luminosity function, CIE 1951 (green)

The dotted line is the most currently accepted luminosity function from 2005

Source: <https://en.wikipedia.org/wiki/Candela#/media/File:Luminosity.png>



Lumen (lm)

- ❖ If a light source emits one candela of luminous intensity uniformly across a solid angle of one steradian, the total luminous flux emitted into that angle is one lumen
 - ❖ $1 \text{ lm} = 1 \text{ cd} \cdot 1 \text{ sr}$
 - ❖ An isotropic one-candela light-source emits a total luminous flux of exactly 4π lumens

Electrical power equivalents for differing lamps^[5]

Minimum light output (lumens)	Electrical power consumption (watts)		
	Incandescent	Compact fluorescent	LED
200	25	3-5	?-?
450	40	9-11	6-8
800	60	13-15	9-12
1,100	75	18-20	13-16
1,600	100	24-28	18-22
2,400	150	30-52	30 ^[6]
3,100	200	49-75	32 ^[7]
4,000	300	75-100	40.5 ^[8]

Reference: [https://en.wikipedia.org/wiki/Lumen_\(unit\)](https://en.wikipedia.org/wiki/Lumen_(unit))



Lux (lx)

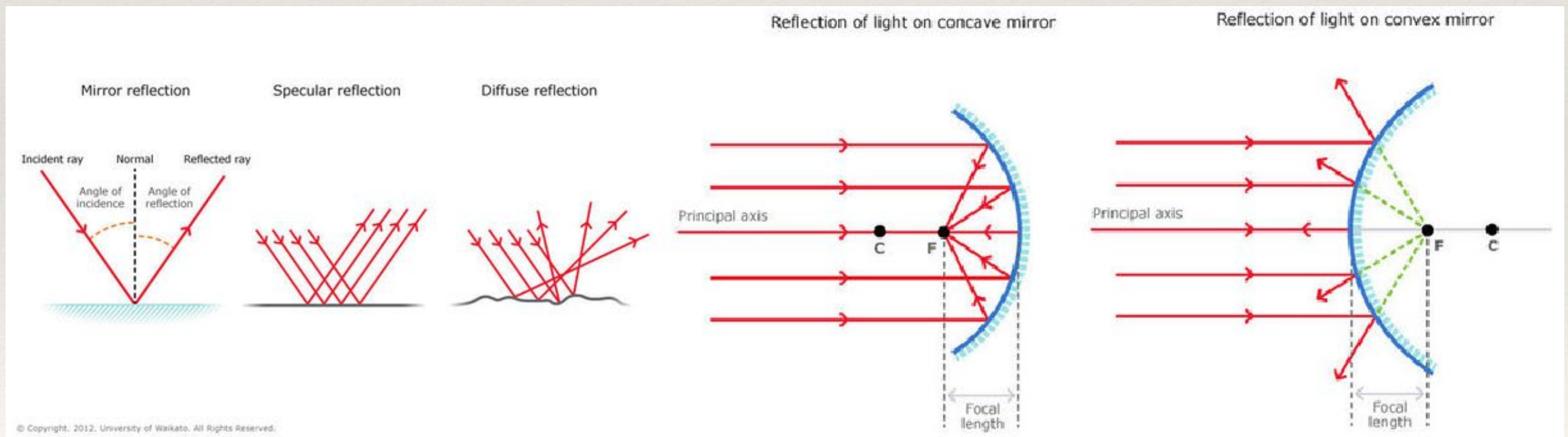
- ❖ The difference between the units lumen and lux is that the lux takes into account the area over which the luminous flux is spread
 - ❖ A flux of 1000 lumens, concentrated into an area of one square metre, lights up that square metre with an illuminance of 1000 lux
 - ❖ The same 1000 lumens, spread out over ten square metres, produces a dimmer illuminance of only 100 lux
 - ❖ Mathematically, $1 \text{ lx} = 1 \text{ lm/m}^2$
- ❖ Examples
 - ❖ A full moon overhead at tropical latitudes emits about 1 lx of light
 - ❖ Office lighting usually ranges from 320 to 500 lx, and TV studio lighting is at 1,000 lx
 - ❖ Inside the visible frequency range, humans can see as little as one photon in the dark
 - ❖ But human's eyes can be open in a desert at noon with the sun exerting up to 130,000 lx. This is an incredible adjustment that current human-made light sensors rarely can match!



Properties of Light

❖ Reflection

- ❖ Incident ray, the reflected ray, and the normal to the surface of the mirror all lie in the same plane
- ❖ The angle of reflection is equal to the angle of incidence .

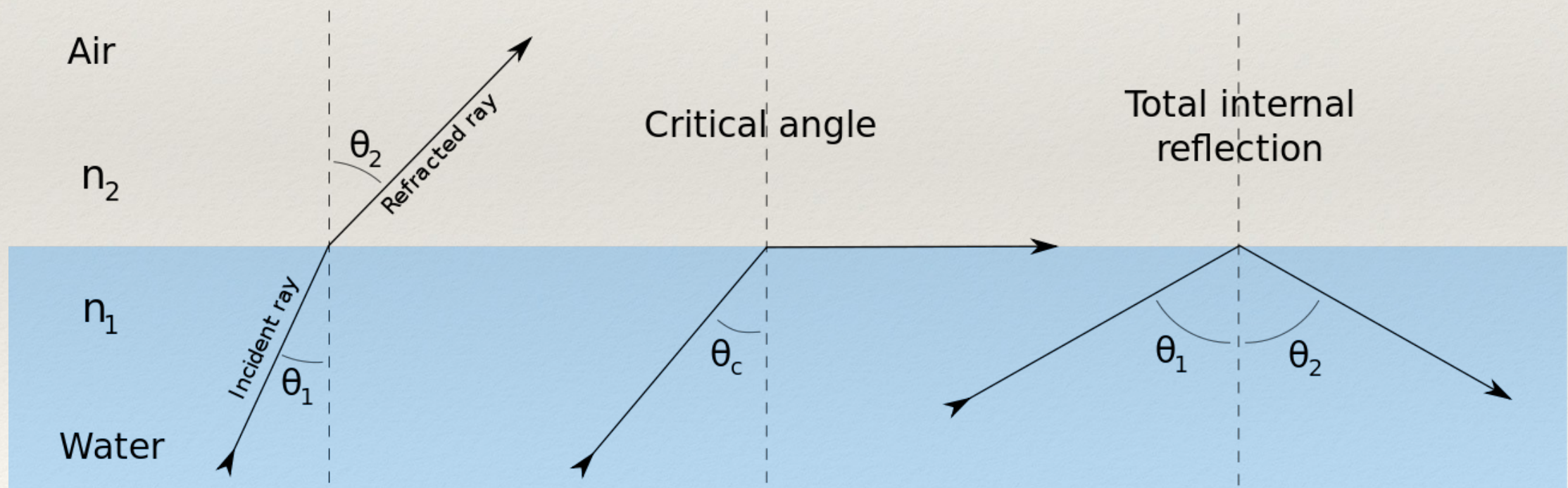


Reference: <http://sciencelearn.org.nz/Contexts/Light-and-Sight/Science-Ideas-and-Concepts/Reflection-of-light>

Properties of Light (Cont'd)

❖ Refraction

- ❖ Refraction is the “bending” of light rays during the transition from one optically transparent material to another
- ❖ The reason for the effect is that the light’s wavelength changes, while the frequency remain unchanged
- ❖ It is responsible for image formation by lenses and eyes



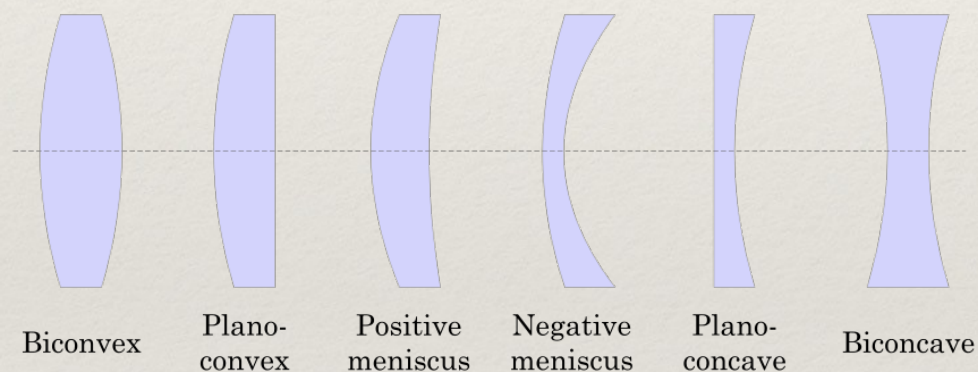
Reference: <https://en.wikipedia.org/wiki/Refraction>



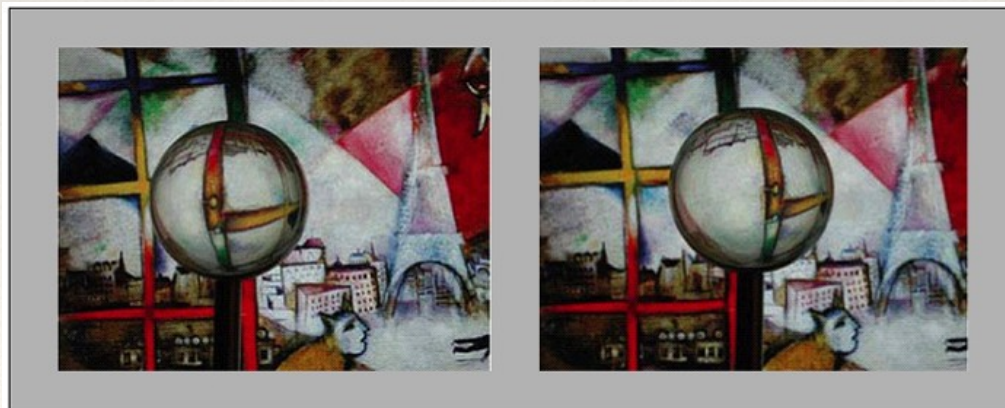
Properties of Light (Cont'd)

❖ Refraction (Cont'd)

- ❖ It is responsible for image formation by lenses and eyes
- ❖ Refraction in transparent objects, and reflection from mirror objects, complicate the detection of objects by computer vision technique



Reference: [https://en.wikipedia.org/wiki/Lens_\(optics\)](https://en.wikipedia.org/wiki/Lens_(optics))



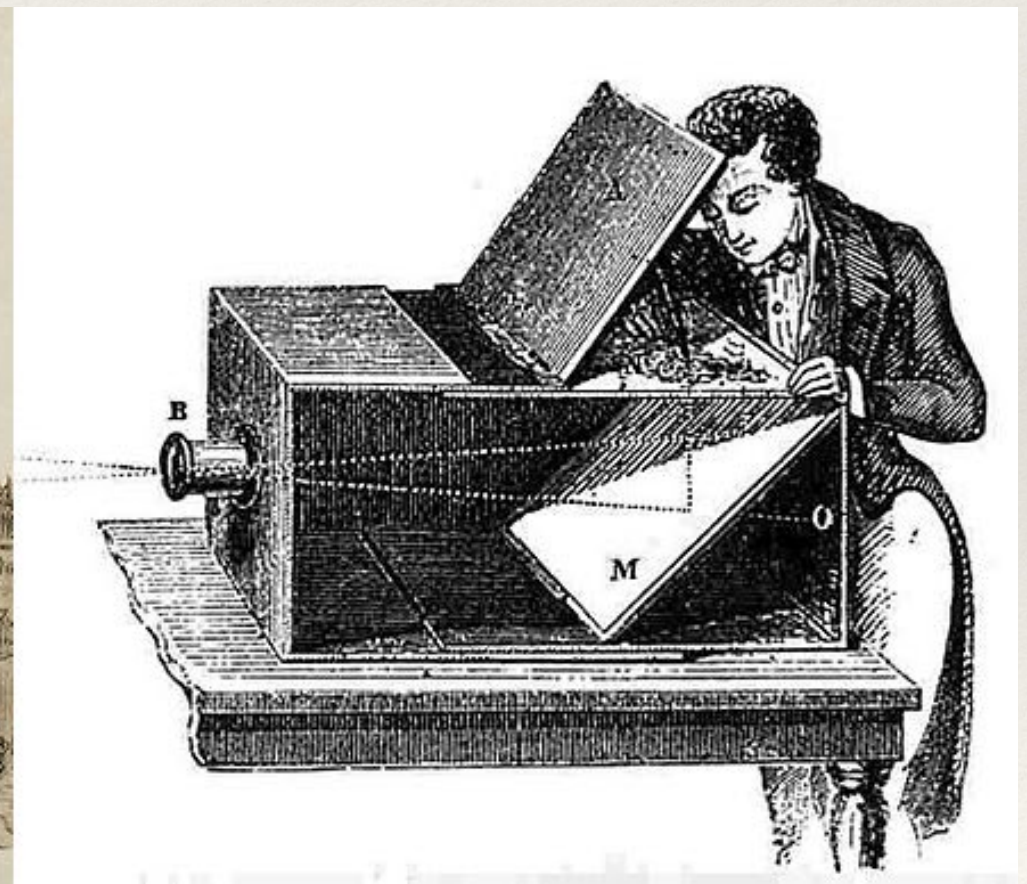
Source: http://www.cs.columbia.edu/CAVE/projects/mo_trans/



Source: http://www.winvow.com/Technology/Technology50_en

Recording Light

- ❖ Camera comes from the Latin camera obscura (“dark chamber”)
 - ❖ Camera obscura could only project light but could not store images



Source: https://en.wikipedia.org/wiki/History_of_the_camera



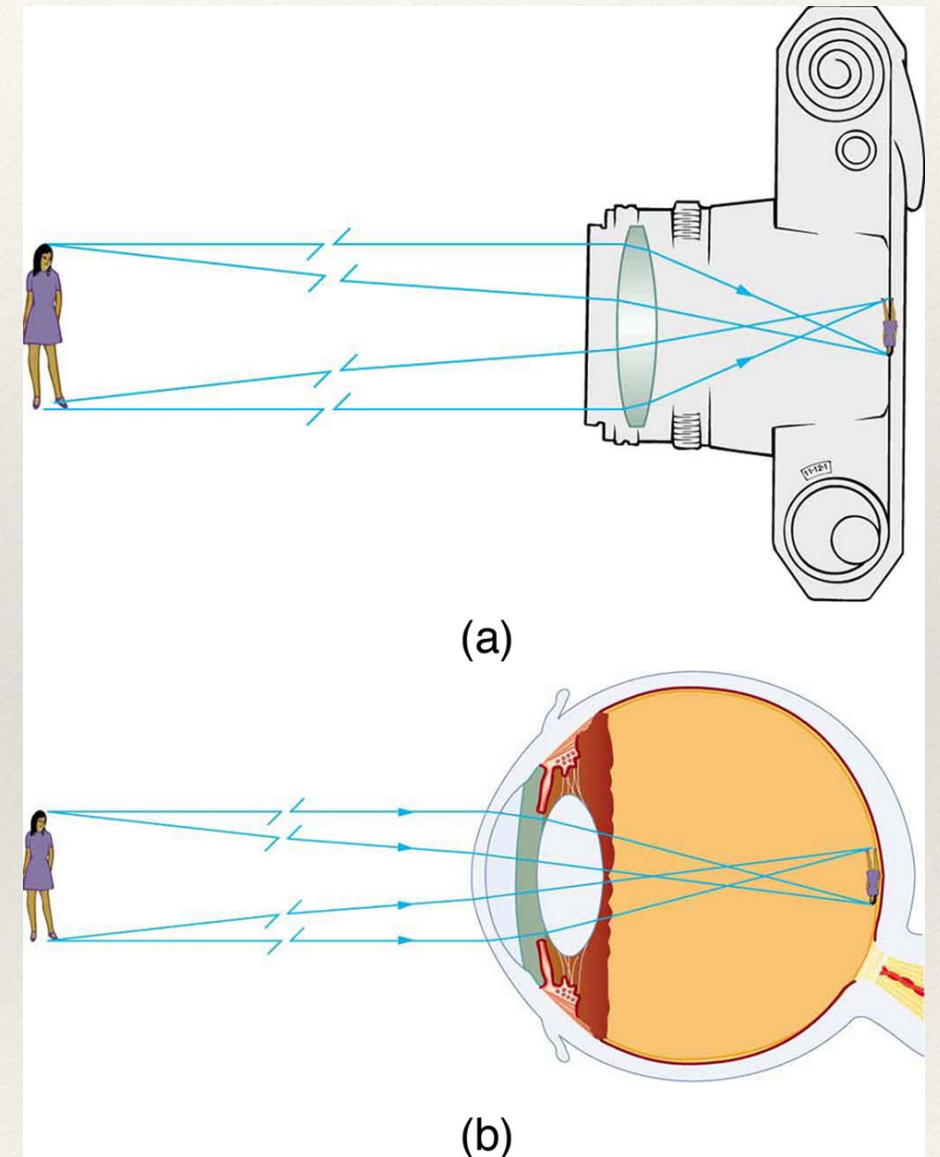
Image Recording

- ❖ Image is projected onto a light-sensitive memory 
- ❖ The first form of light-sensitive memory was paper coated with silver chloride, which darkened upon exposure to light
- ❖ Subsequent advances in technology lead to the invention of photographic film
- ❖ Now, all these light-sensitive medium are replaced by photosensitive electronics that can record images in a digital format



Image Formation

- ❖ The image formation process consists of two parts:
 - ❖ The geometry of image formation, which determines where in the image plane the projection of a point in the scene will be located
 - ❖ The physics of light, which determines the brightness of a point in the image plane as a function of scene illumination and surface properties

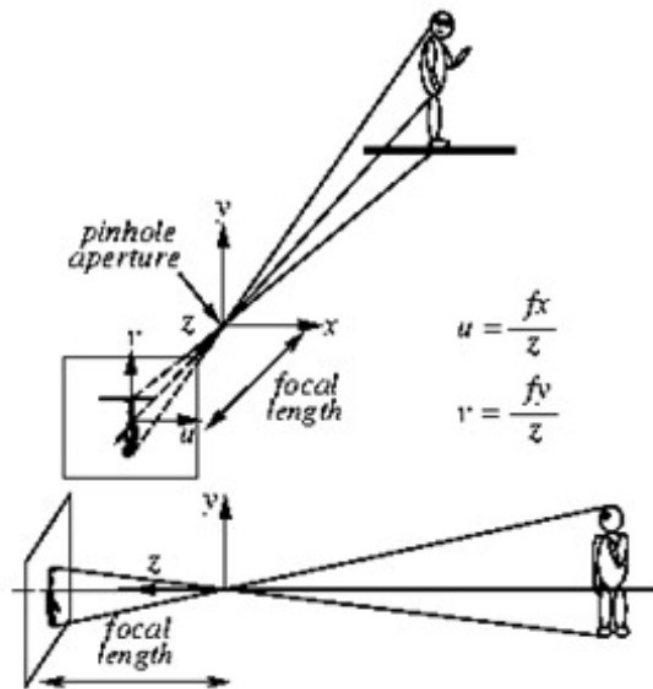


Source: http://cnx.org/resources/8d75c2ba3ba78a9d4ab6862595fb14e9ff1dc903/Figure%2026_06_08.jpg



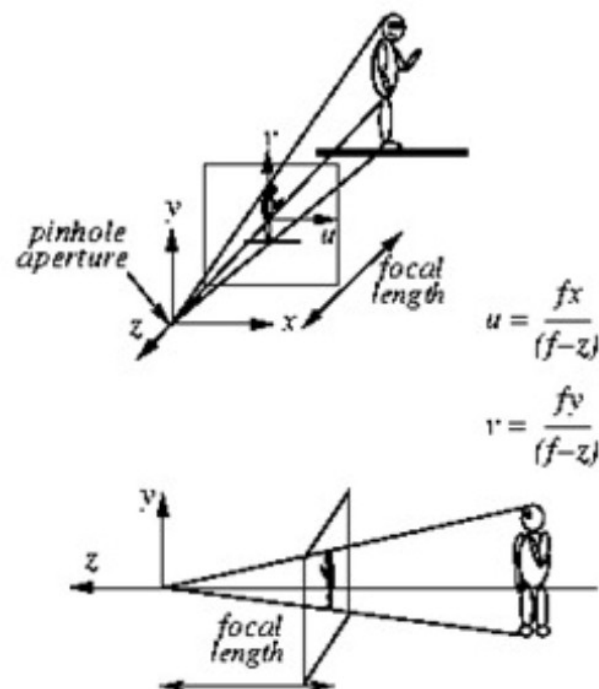
Geometry of Image Formation

- ❖ Basic model for the projection of points in the scene onto the image plane
- ❖ Camera converts a 3D world to a two dimensional image

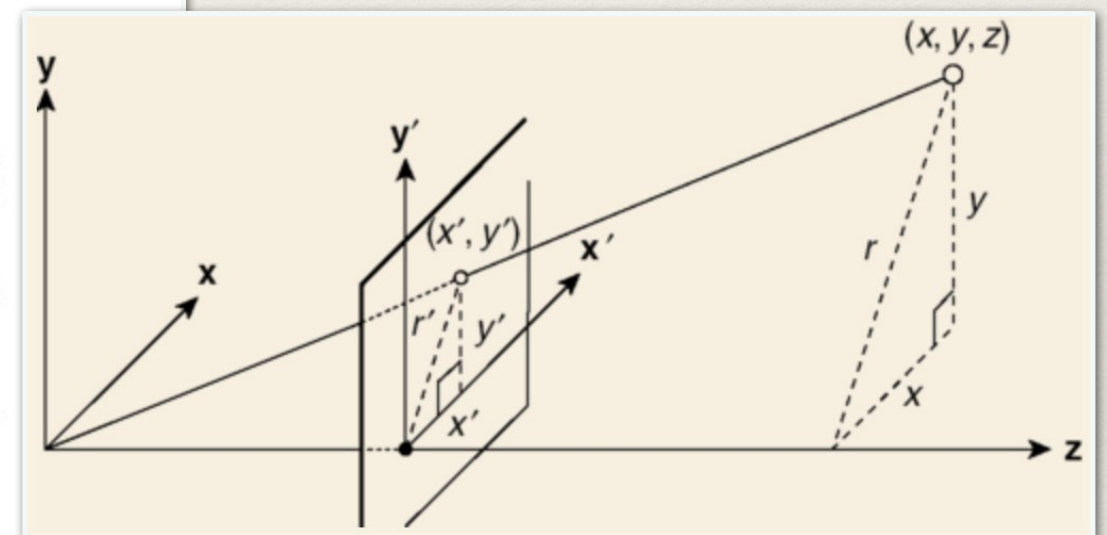


Perspective projection geometry.

Source: <http://slideplayer.com/slide/4960863/>



Mathematically equivalent non-inverting geometry.

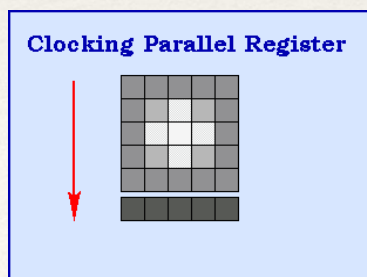
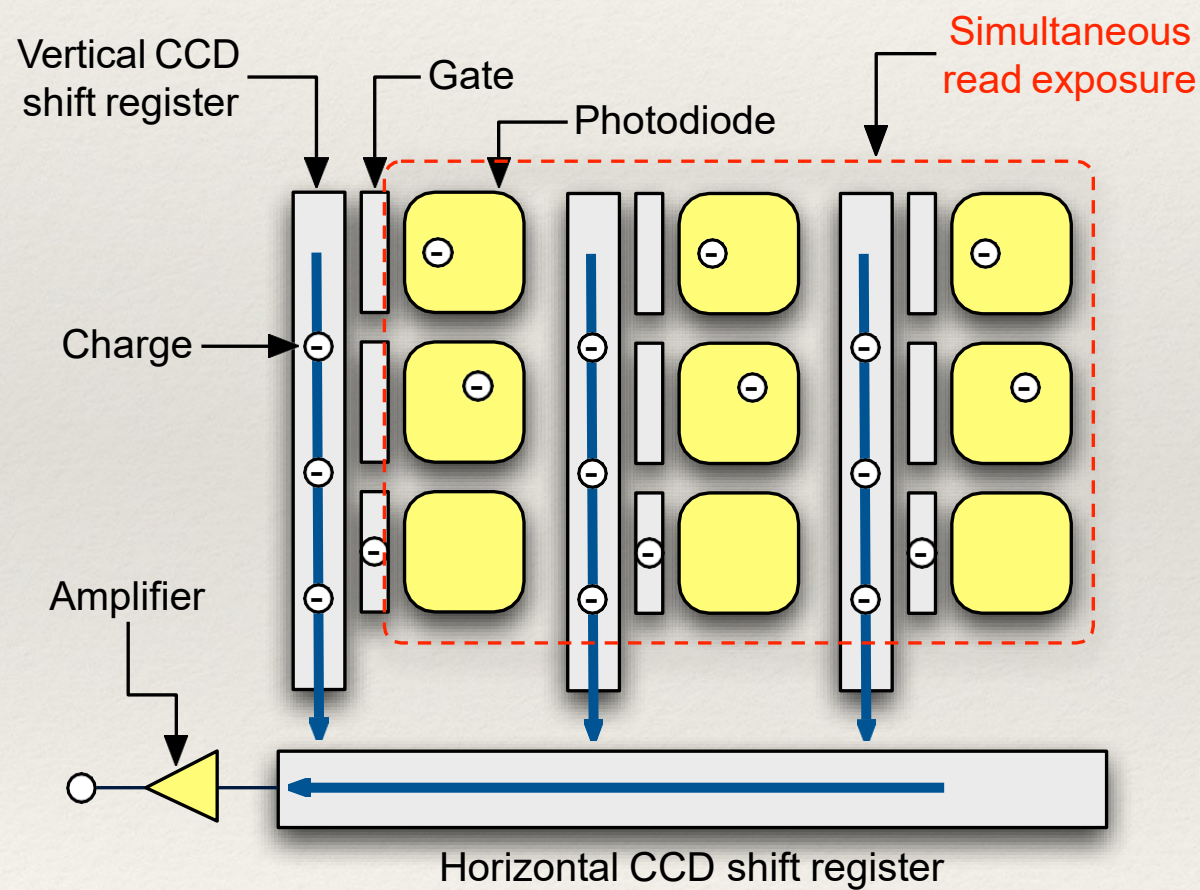


Camera Sensors

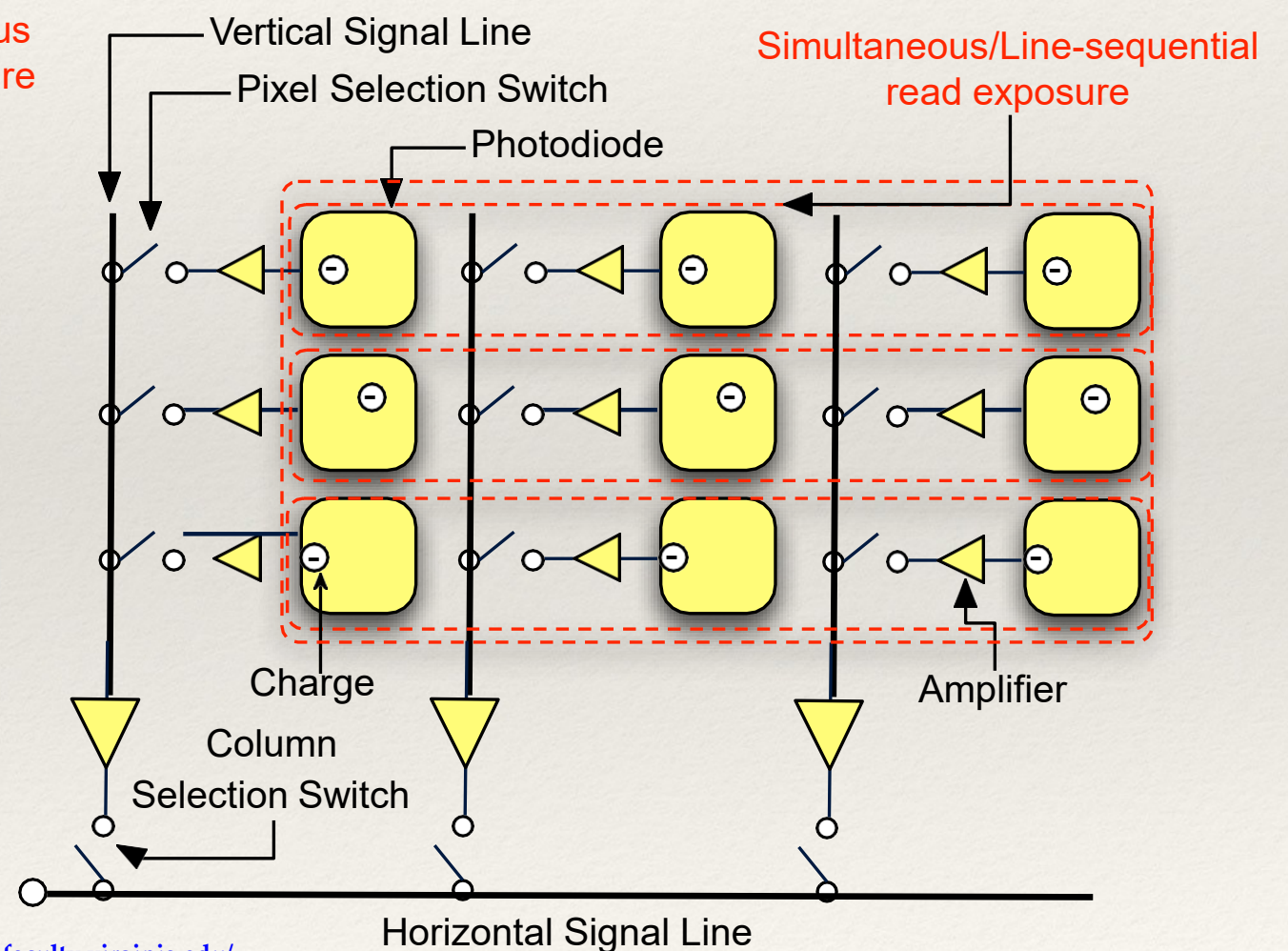
❖ CCD or CMOS

- ❖ Both employed pixelated metal oxide semiconductors as sensing elements

CCD Sensor Structure



CMOS Sensor Structure



Source: http://www.faculty.virginia.edu/ASTR5110/lectures/detectors/detectors_readout.html

Camera Sensors (Cont'd)

❖ CCD or CMOS

- ❖ Megapixel Camera are more inclined towards CMOS sensors
- ❖ CCD sensors still deliver better performance for scene with low-lighting

	CCD	CMOS
Pixel Transfers	Shift Register Transfer	Direct Pixel Address
Charge-to-Voltage Conversion	After pixel readout	In-Pixel
Exposure	Single Frame/Field Exposure	Single Frame/Line Sequential Exposure
Sensitivity	Higher	Lower
Control Circuit	Off-Sensor	Sensor Integrated



Resolution

- ❖ Digital cameras is defined by number of photoelectric sensors
- ❖ Each sensor creates one picture element, a.k.a. pixel

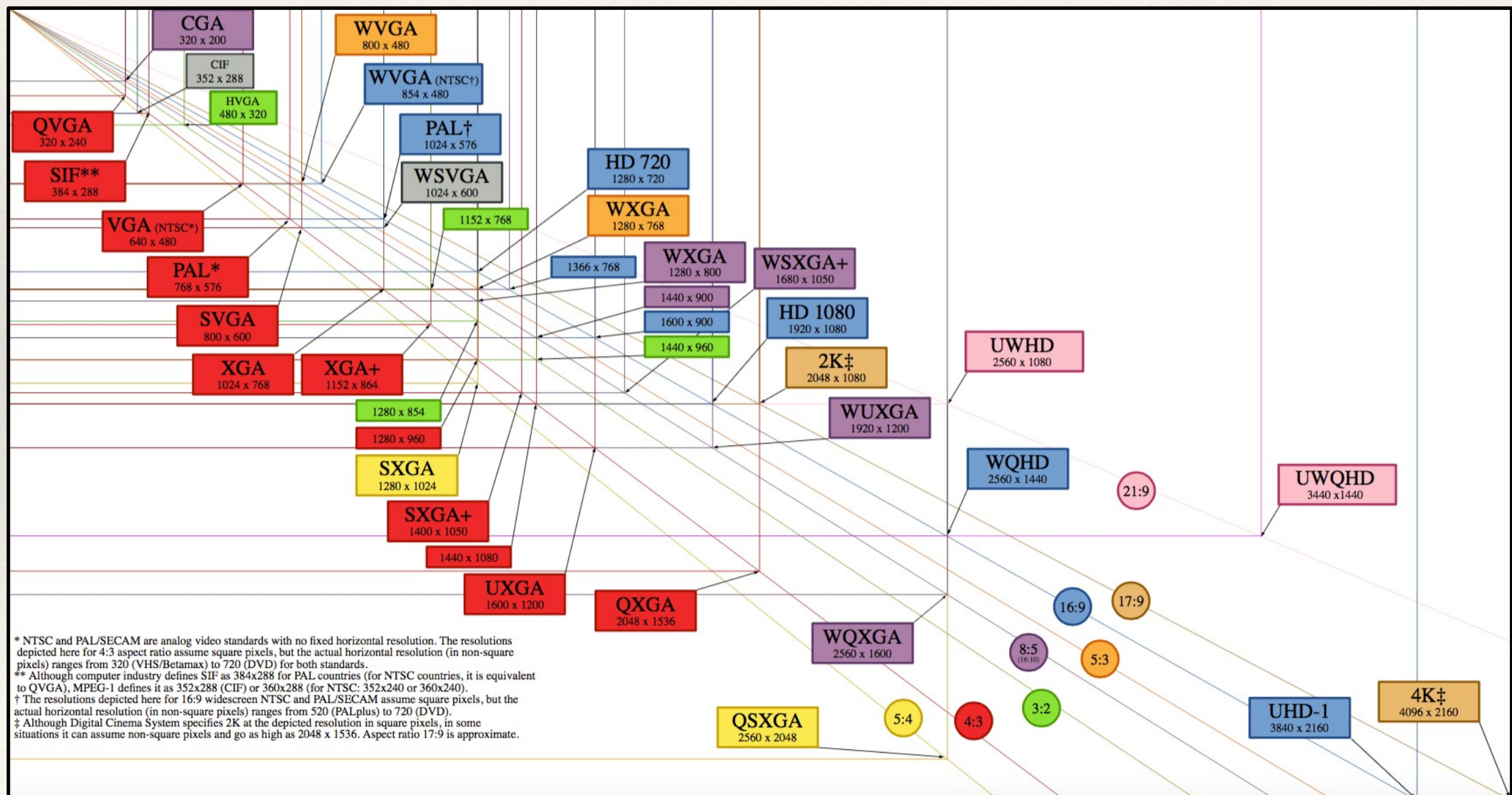


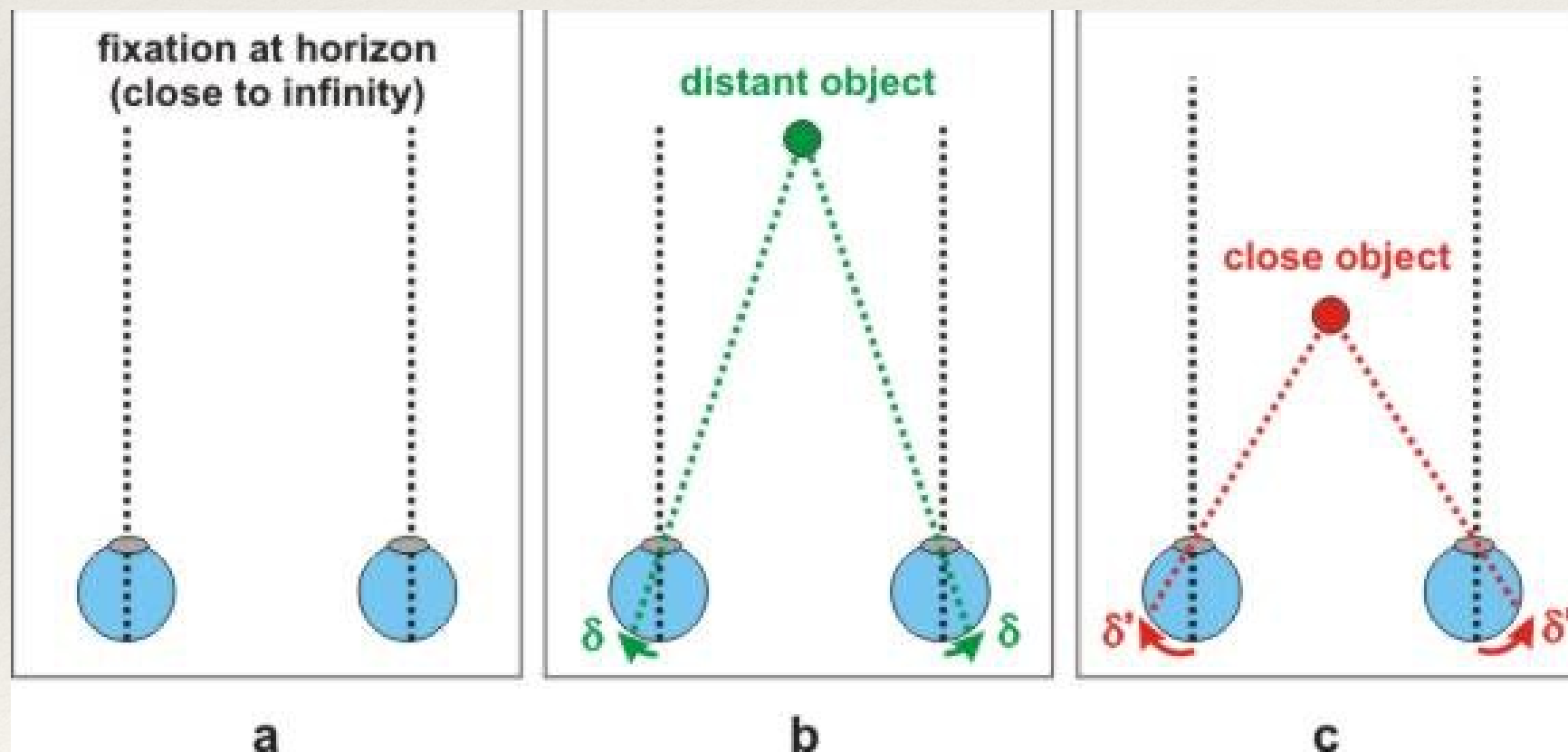
Photo & Video Cameras

- ❖ Photo Camera takes still images
- ❖ Video Camera record sequence of images
 - ❖ FPS - Number of **F**rames captured **P**er **S**econd
 - ❖ These images are compressed and recorded as one single video file
- ❖ Photo Camera's resolution is up to 32 Mega Pixels
- ❖ Video Camera's resolution is up to 16 Mega Pixels



Stereo Camera

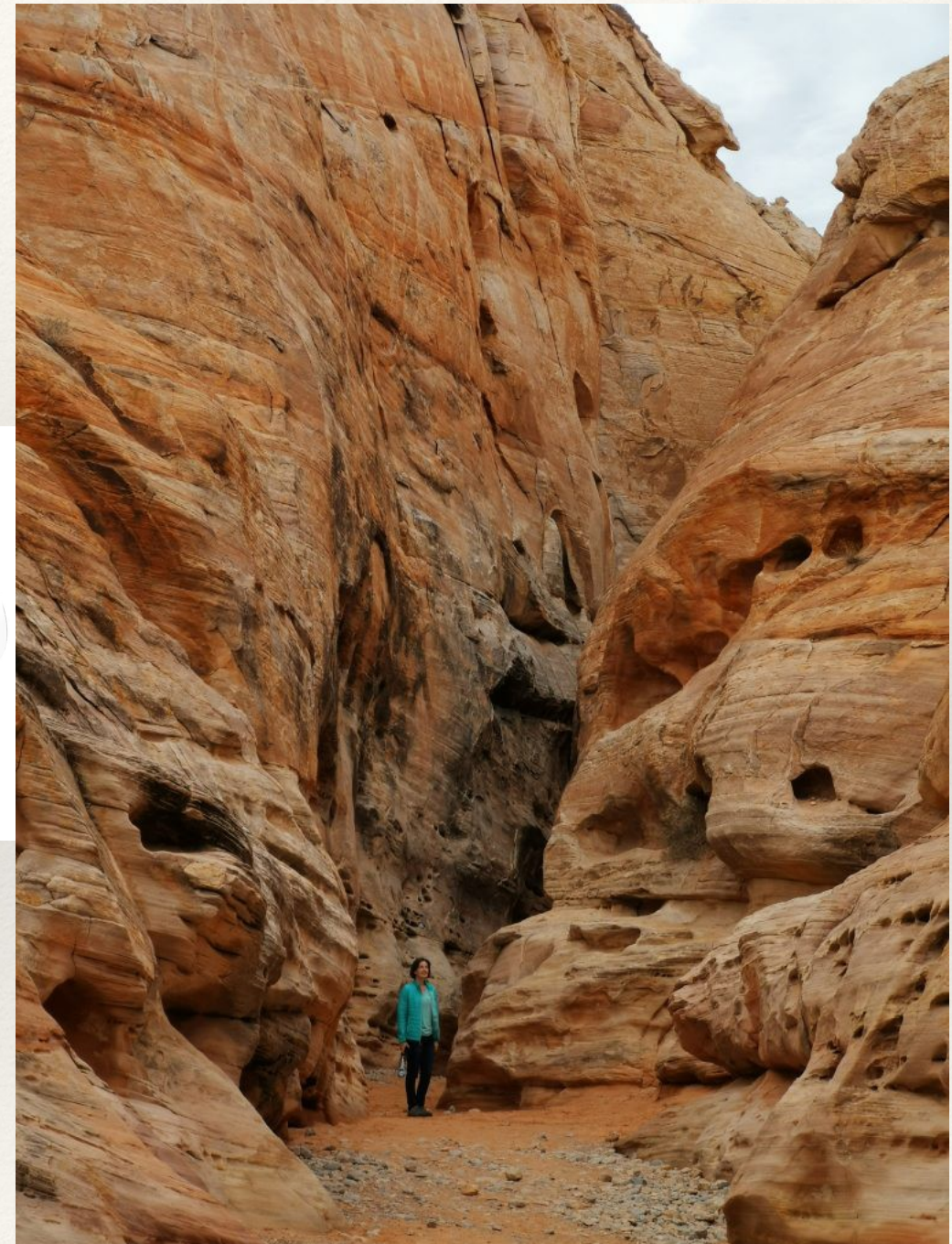
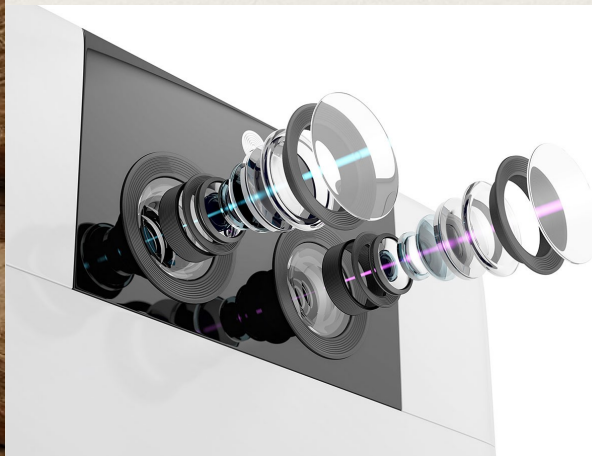
- ❖ Stereo Camera can capture depth information, which is essential for capturing 3D images



Source: <http://www.pc.rhul.ac.uk/staff/J.Zanker/PS1061/L4/stereo.jpg>



Multiple Cameras on Mobile Phone



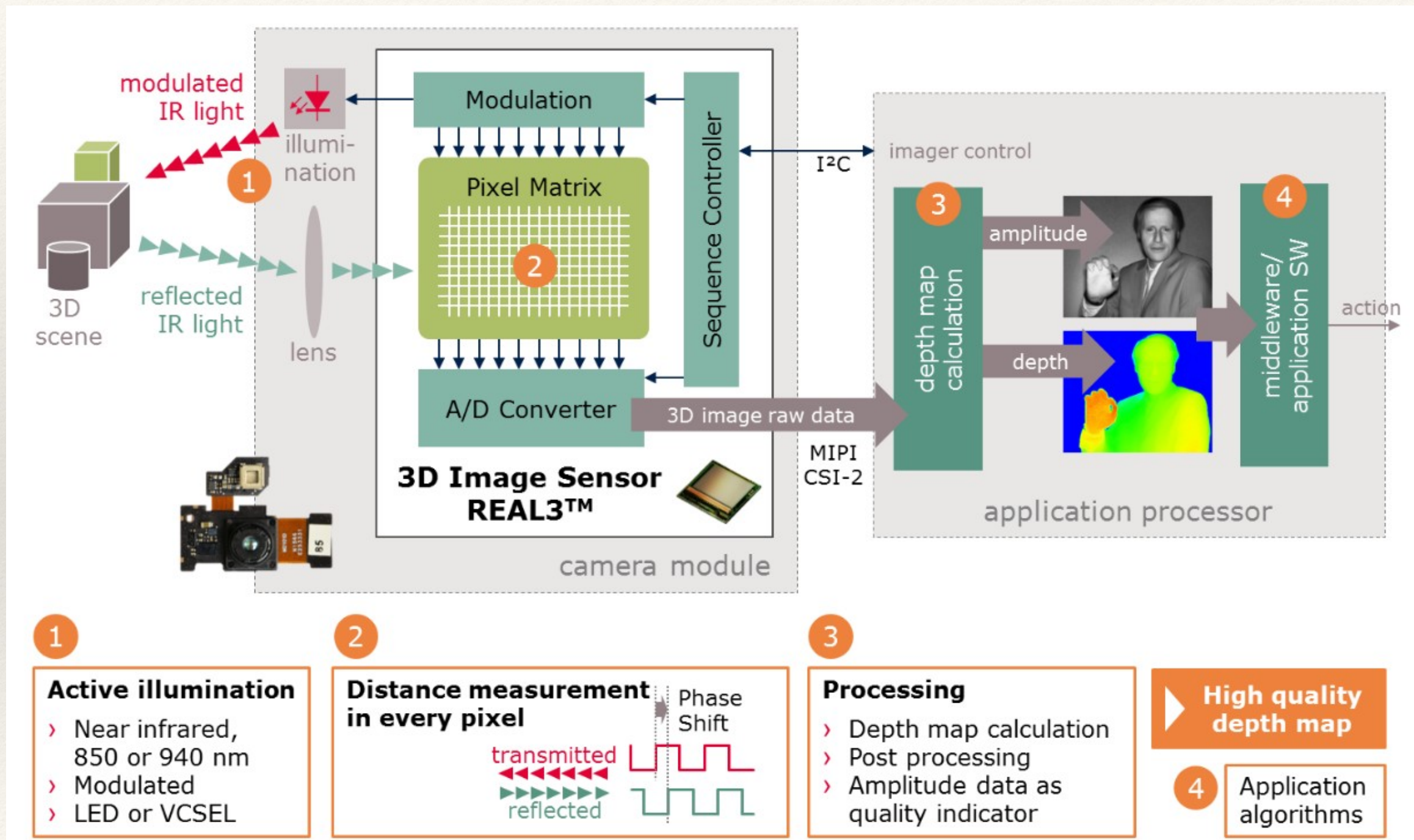
Ref: <https://www.dxomark.com/multi-camera-smartphones-benefits-and-challenges/>



Multiple Cameras on Mobile Phone

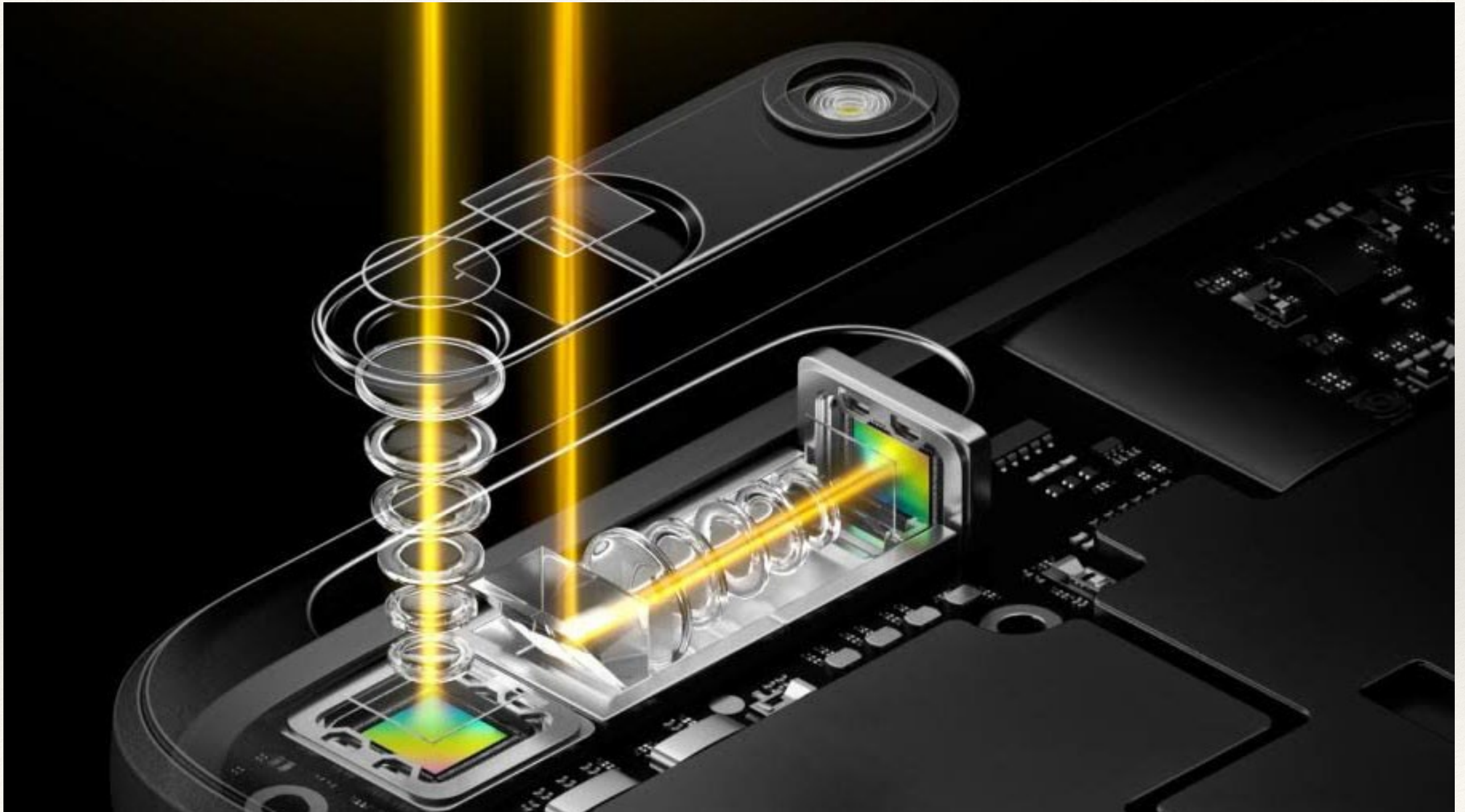


3D Time-of-Flight (ToF) Sensor



Source: <https://www.infineon.com/cms/en/product/sensor/3d-image-sensor-real3/>

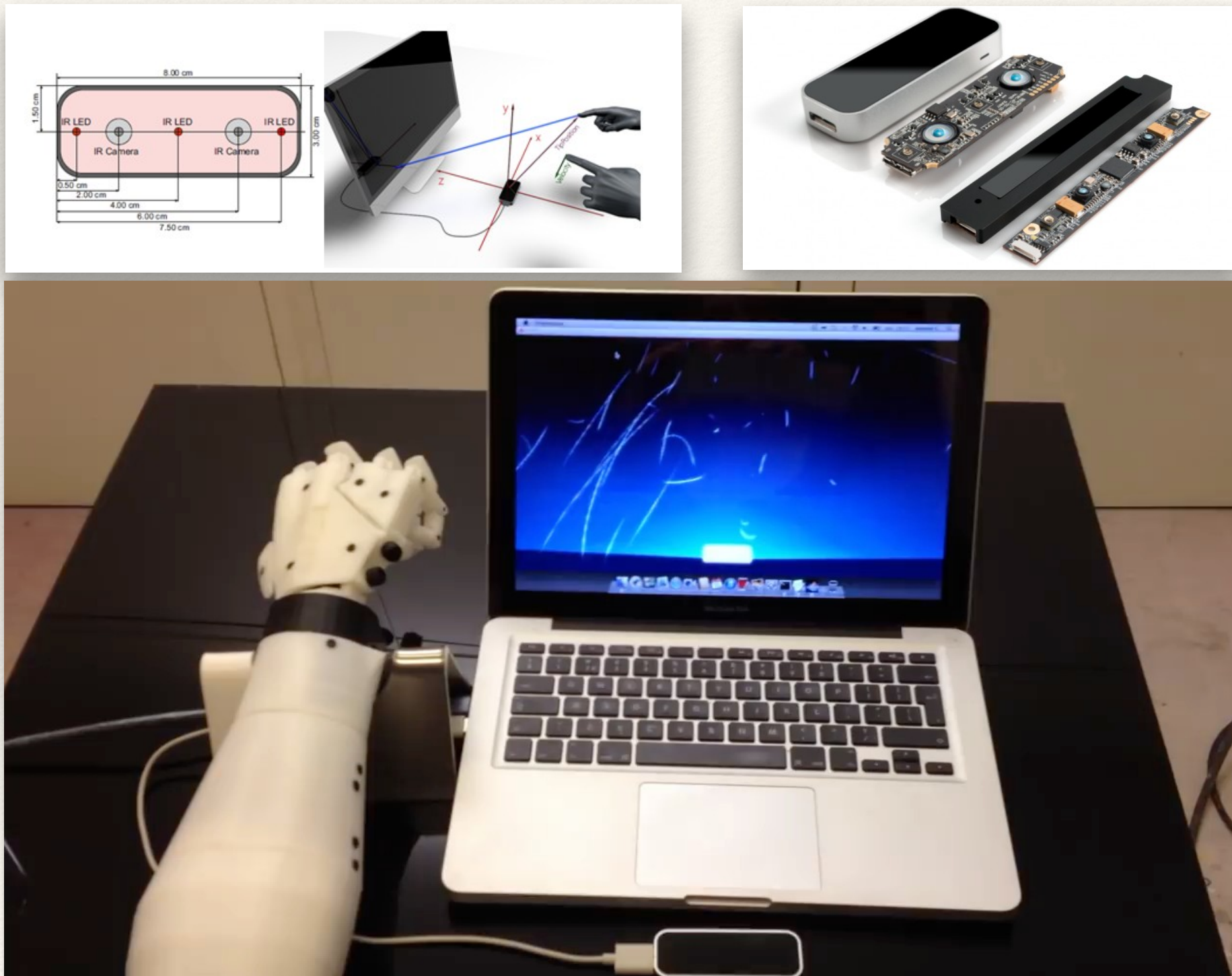
Folded Optics



Ref: <https://www.dxomark.com/multi-camera-smartphones-benefits-and-challenges/>



Leap Motion Sensor



Source: <https://www.youtube.com/watch?v=GOTRXM4O9L0>



Reproducing Light

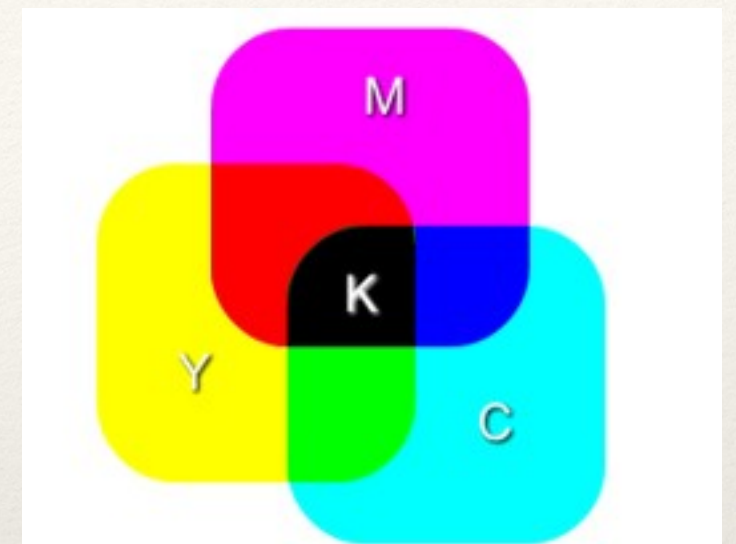
❖ Two main methods

❖ Subtractive

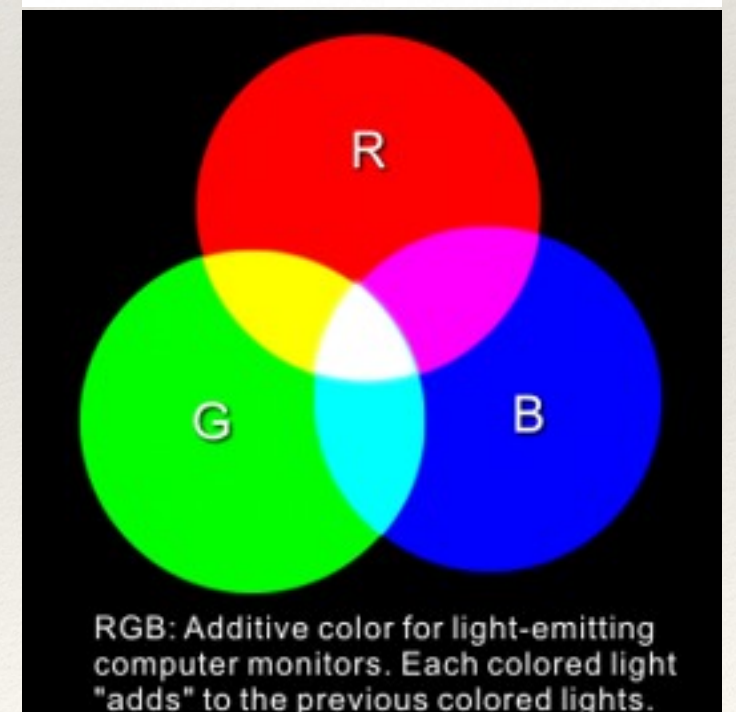
- ❖ Methods rely on intensity variations of the reflection of ambient light and do not work when no light is present. Paper, for example, reflects patterns differently once it has been modified by ink or toner

❖ Additive

- ❖ Methods work with active light sources that are mixed together. The most common example is the display in a TV or monitor



CMYK: subtractive colors for printer. Each color added to the first color blocks the reflection of color, thus 'subtracts' color.

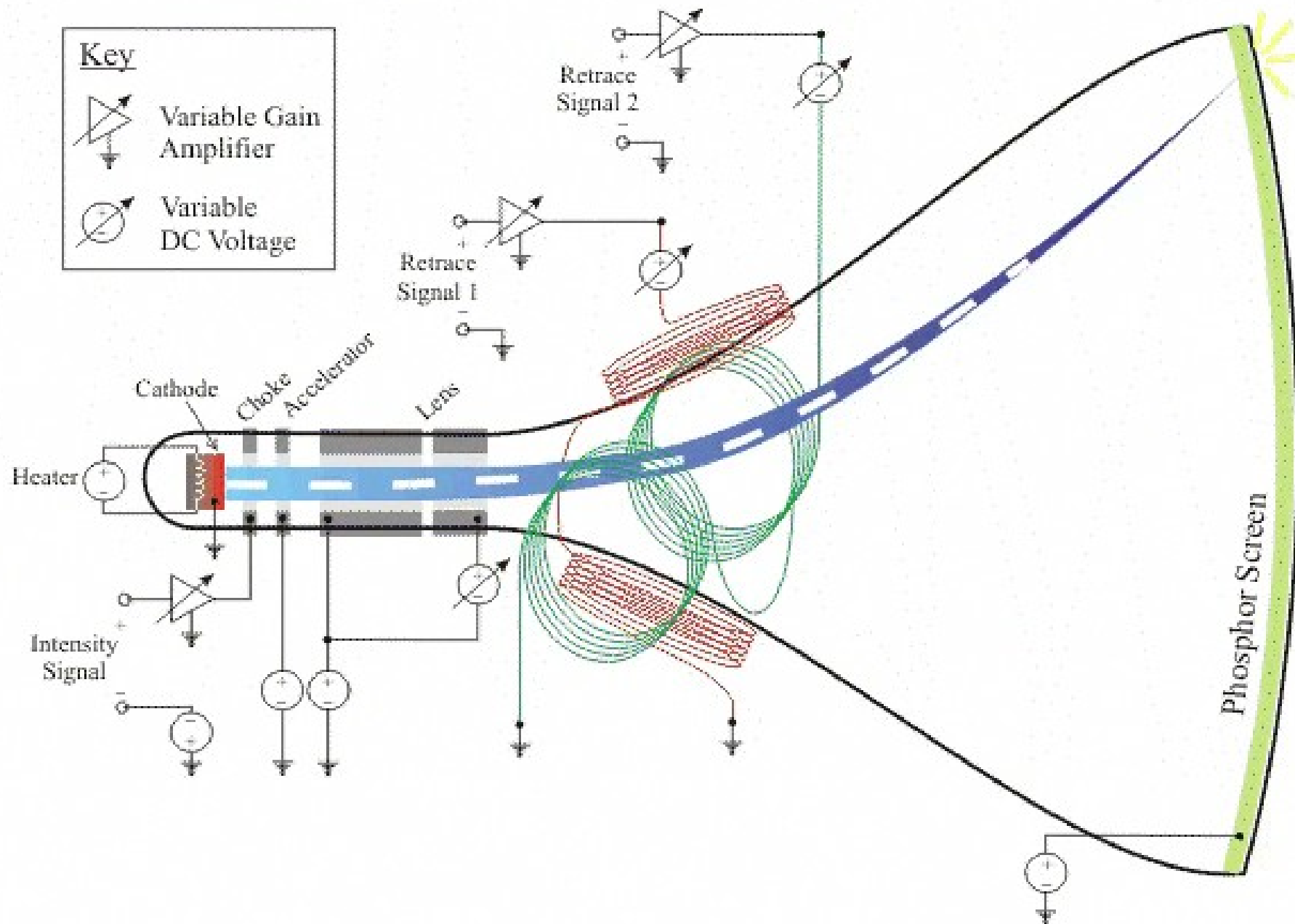


RGB: Additive color for light-emitting computer monitors. Each colored light "adds" to the previous colored lights.

Source: <http://www.burleighprint.com.au/wp-content/uploads/2015/02/cmyk-rgb.jpg>

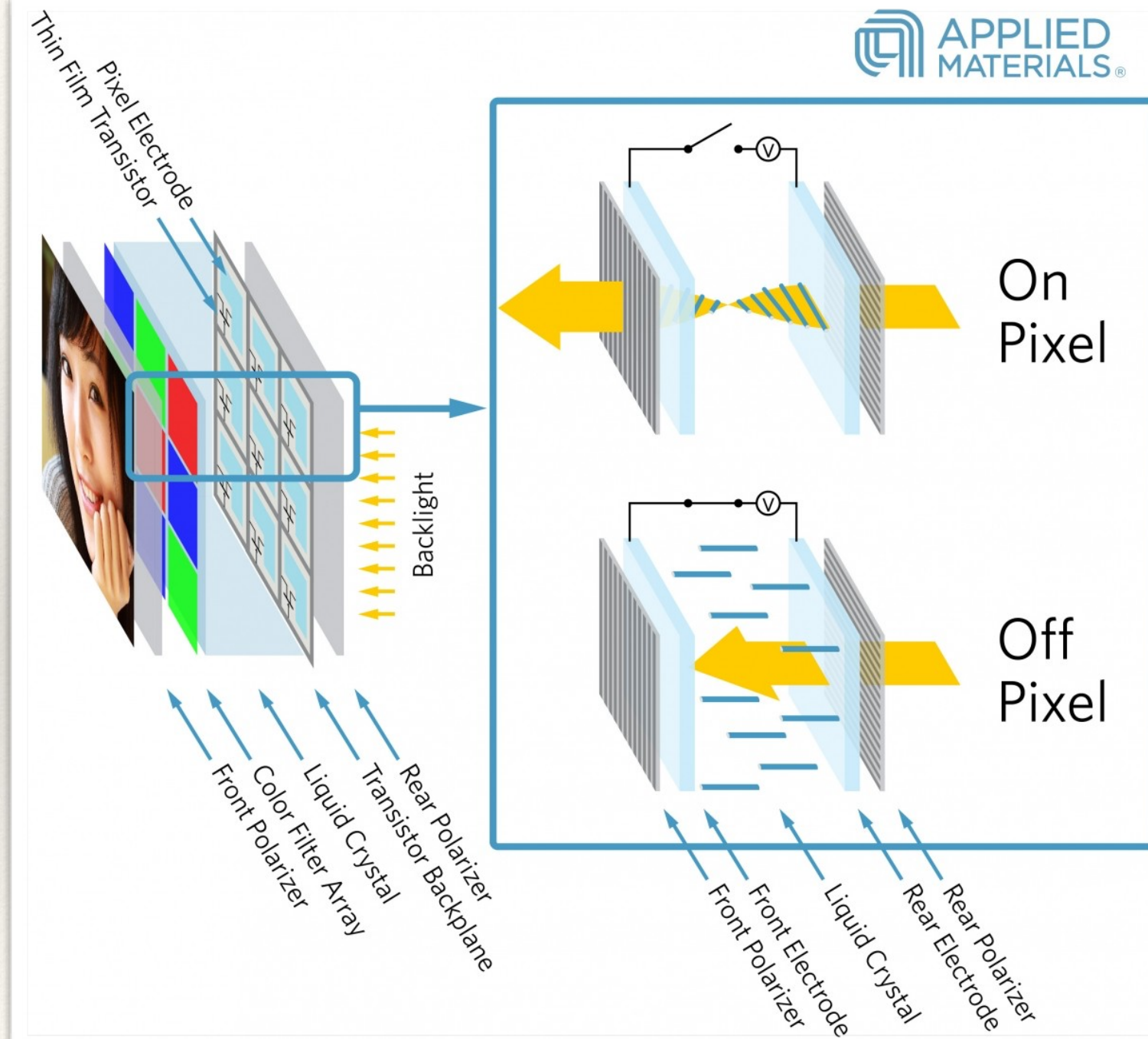


CRT Monitor

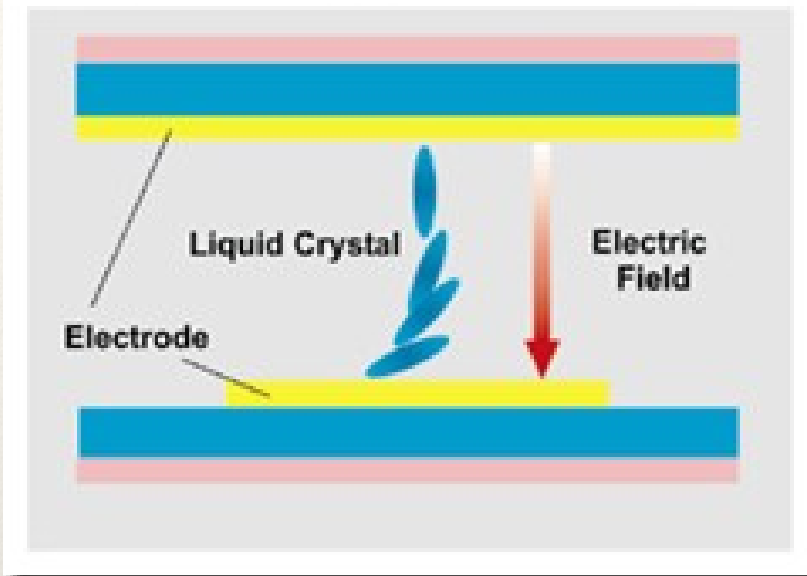


*D.A. de Wolf. *Principles of Electron Optics*. Wiley and Sons. New York. 1991

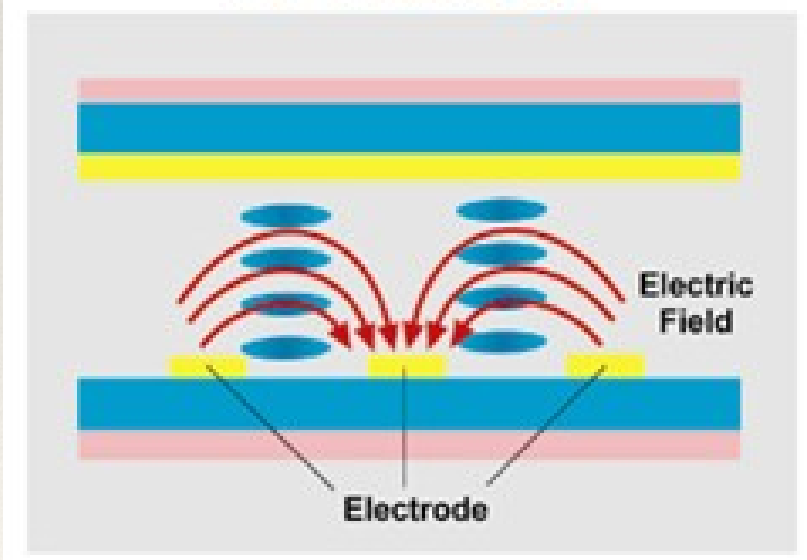
LCD Monitor



Conventional TN Process



IPS Technology

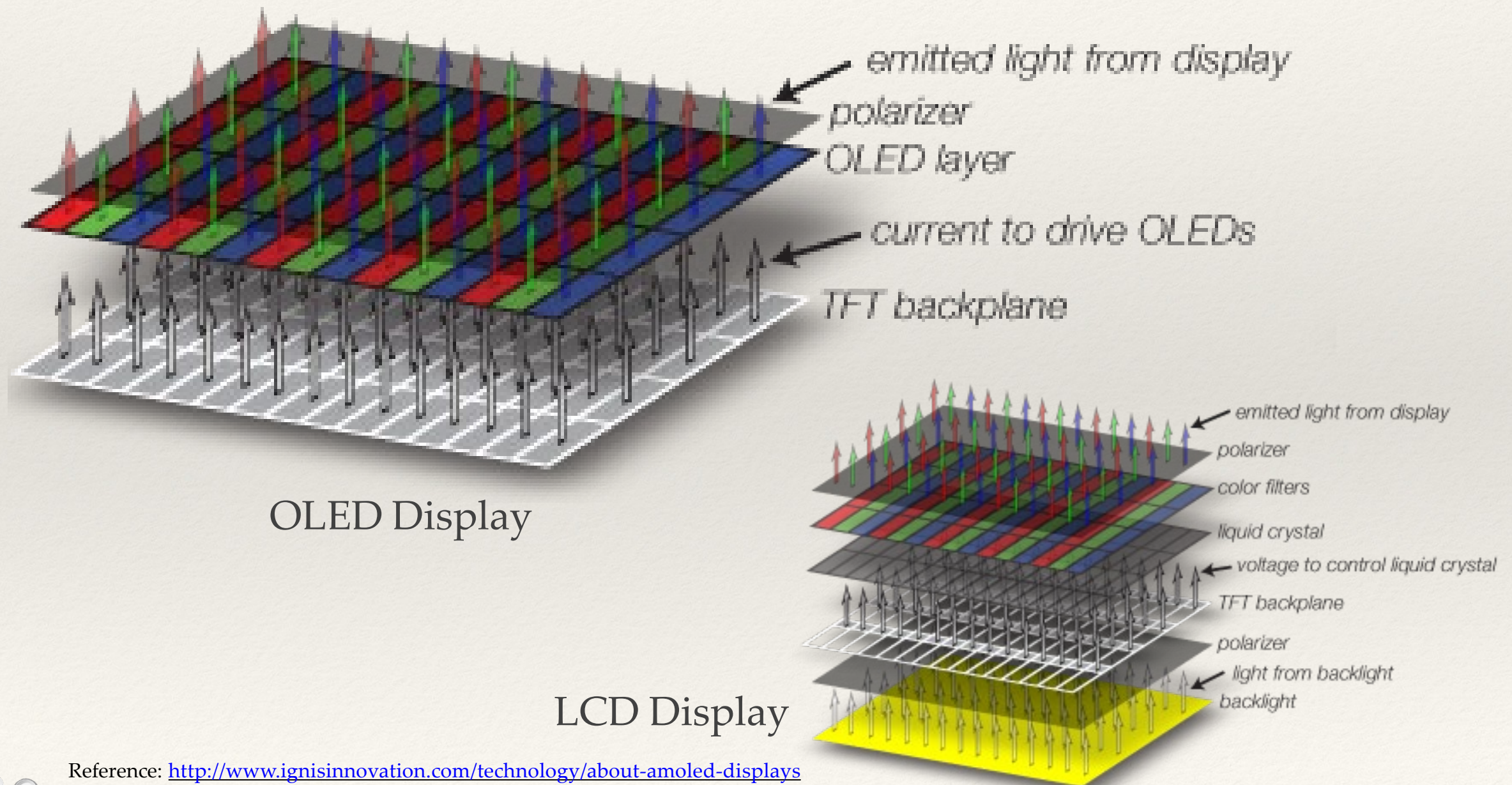


Source:

1. <http://www.displayalliance.com/news/the-flat-panel-display-news-source-3202012.html>
2. <https://splurgebook.wordpress.com/2010/01/31/ips-display-technology-whats-the-big-deal/>

OLED Display

❖ AMOLED - Active Matrix Organic Light Emitting Diode



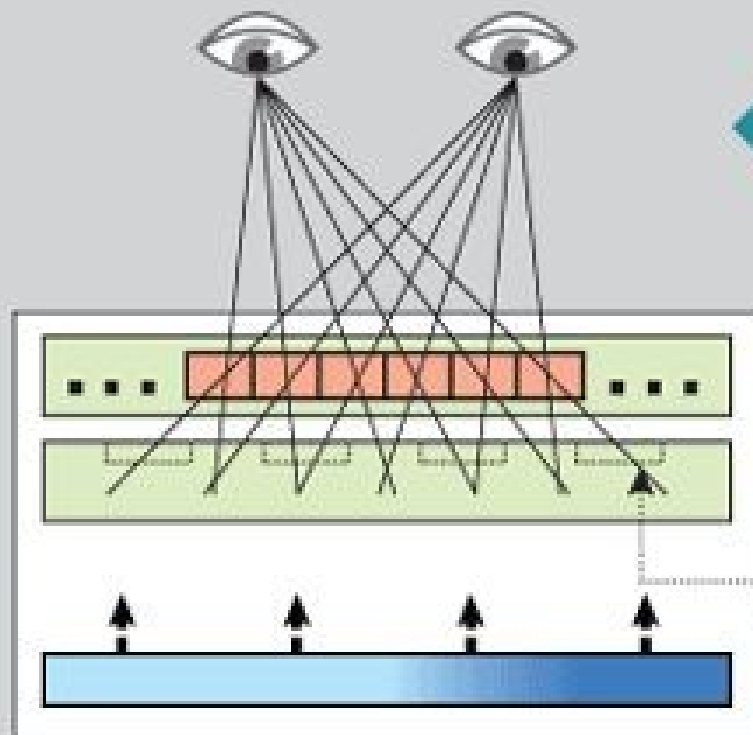
Reference: <http://www.ignisinnovation.com/technology/about-amoled-displays>

Autostereoscopic Display

- ❖ The major disadvantage with this method is that you will have to sit at certain angles called sweet spots to get the best view. If you sit away from the prescribed angle, the display images appear to be blurred

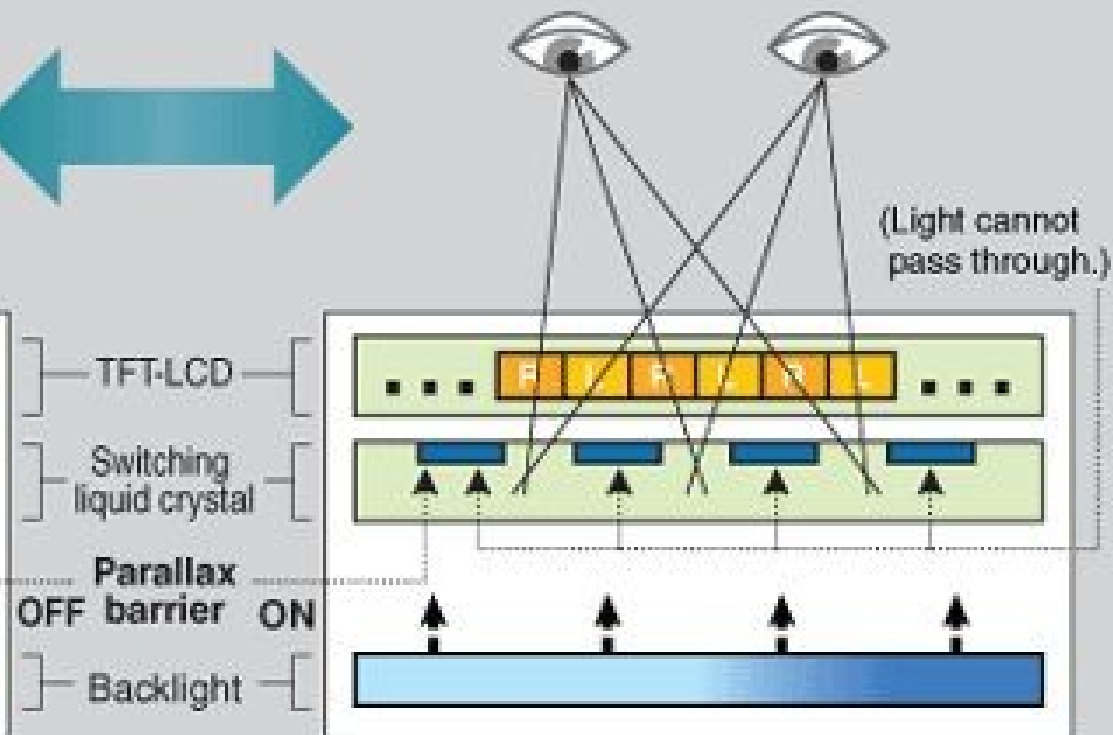
2D display mode

The switching liquid crystal is controlled so as to make the parallax barrier transparent, allowing all the light to pass through. This causes the same image to be seen by the right and left eyes, resulting in a two-dimensional display.



3D display mode

A parallax barrier is created by controlling the switching liquid crystal, thereby separating the light into two images. This causes different images to be seen by the right and left eyes, creating a sense of depth.



3D Hologram

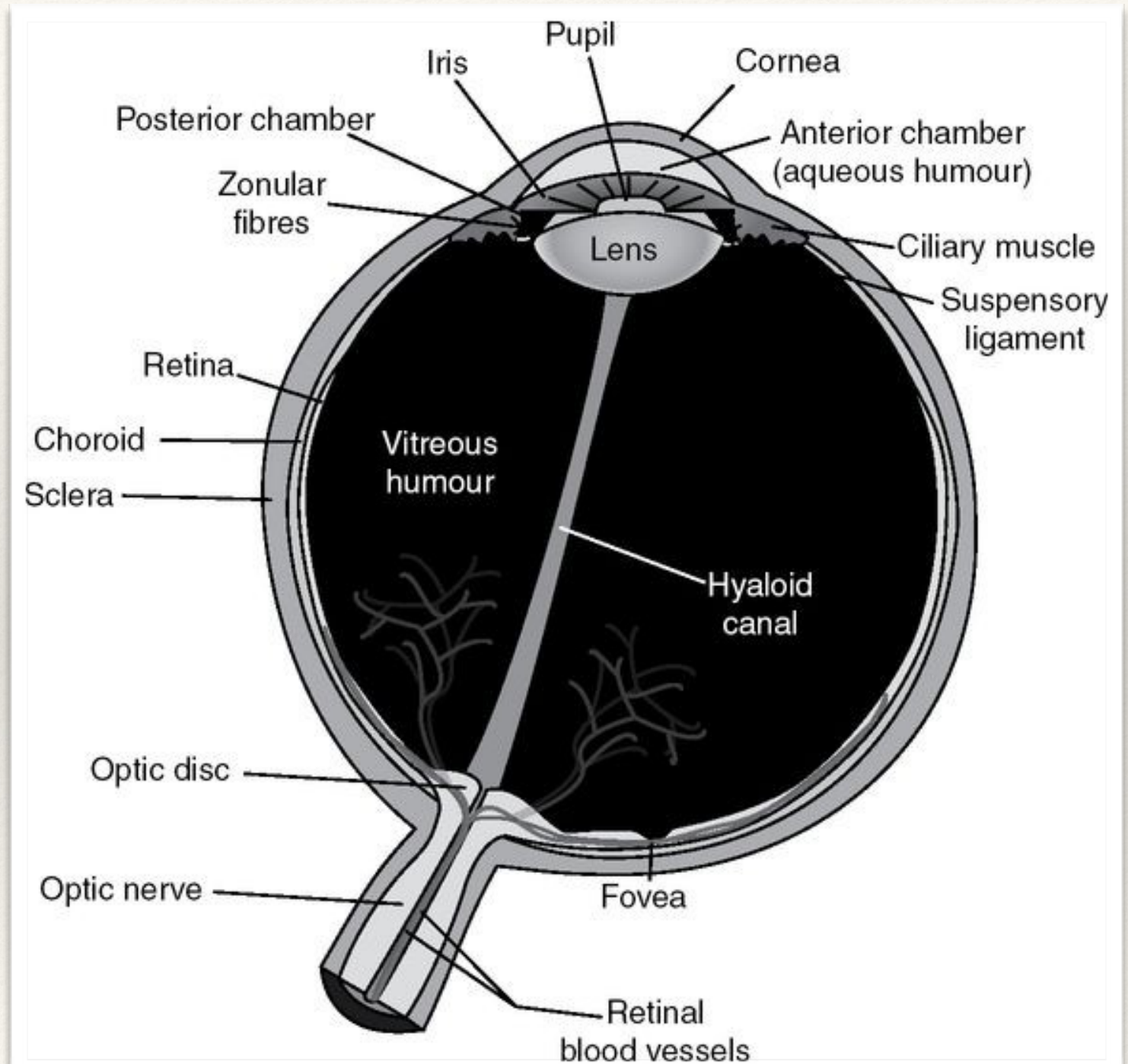


 Gatebox



Perception of Light

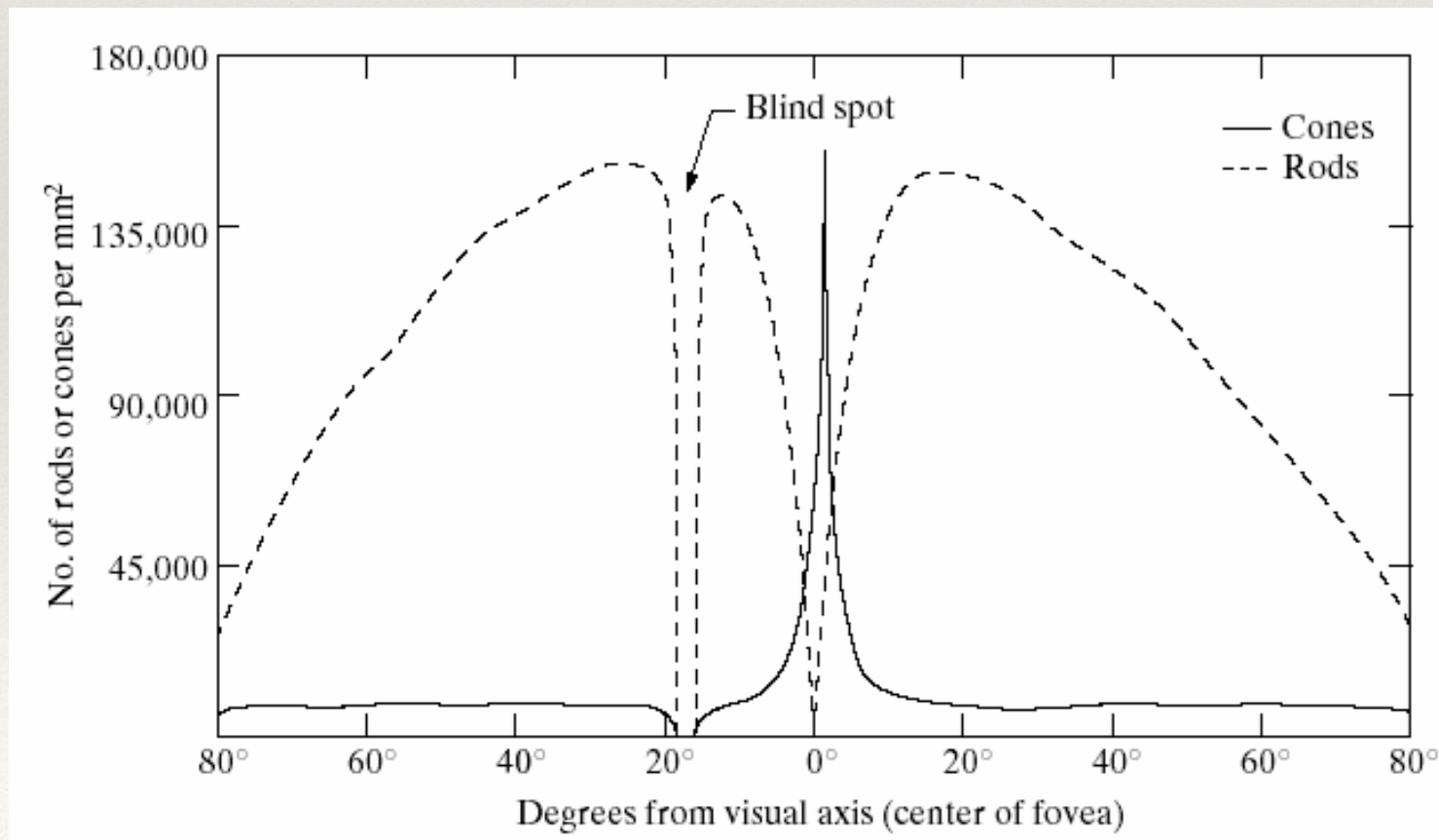
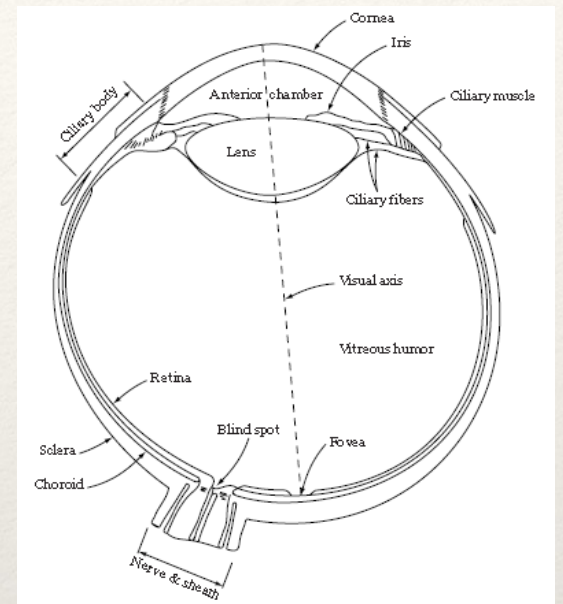
- ❖ Basic structure of an eyeball:
 - ❖ *iris*: controls the amounts of light entering the eye
 - ❖ *lens*: focuses the image onto the retina
 - ❖ *retina*: covered by light receptors (i.e., *cones & rods*) that convert the image into chemical and nervous signals that are transmitted to the brain via the *optical nerves*



Photopic (Bright-Light) Vision

❖ Cones in the retina

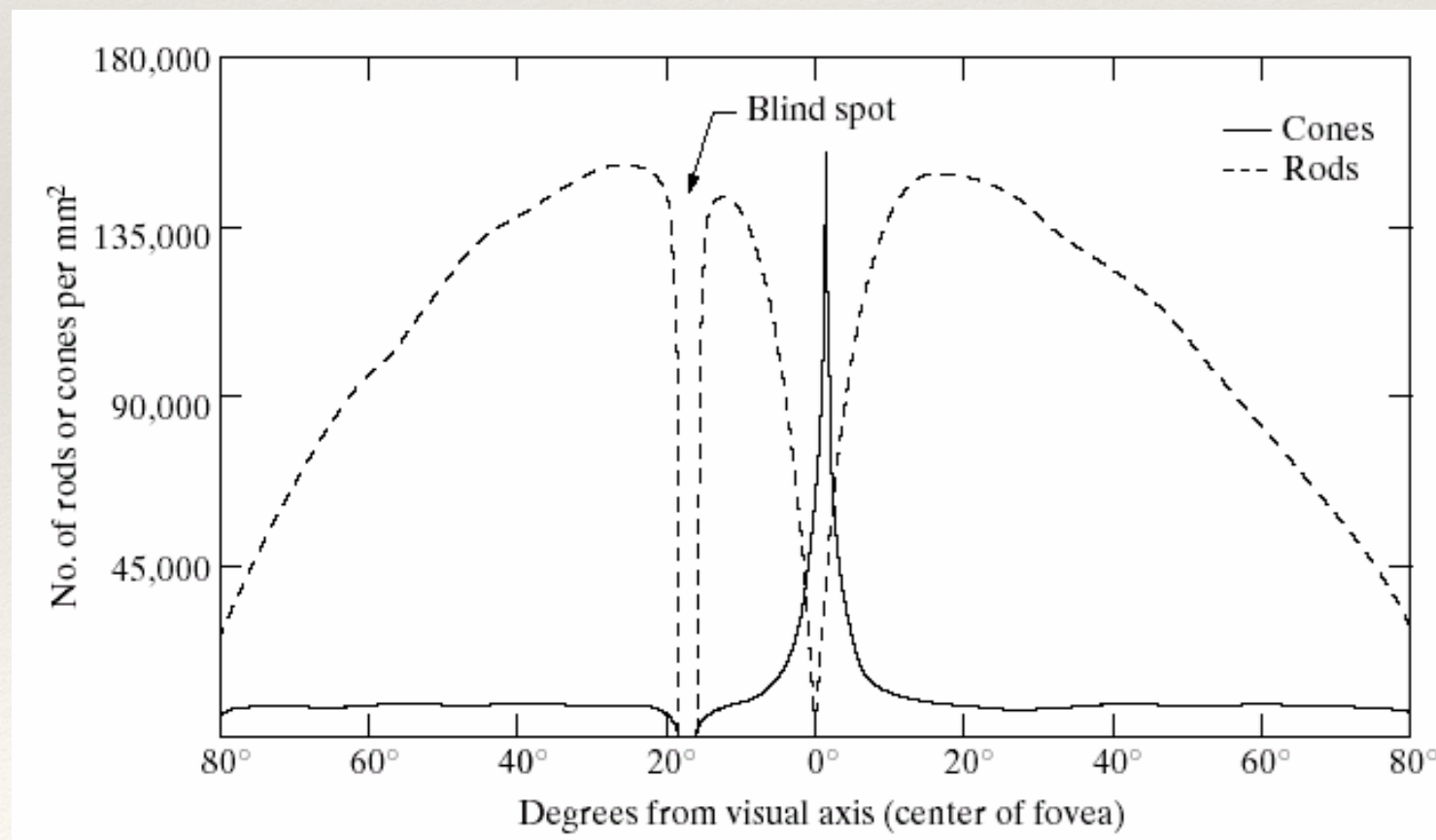
- ❖ Number between 6 and 7 millions
- ❖ Located primarily in the central portion of the retina called the *fovea*
- ❖ Each one is connected to its own nerve end
- ❖ Highly sensitive to color, and give fine details



Scotopic (Dim-Light) Vision

❖ Rods in the retina

- ❖ Number between 75 and 150 millions
- ❖ Distributed over the retinal surface
- ❖ Several rods are connected to a single nerve
- ❖ Serve to give a general, overall picture of the field of view



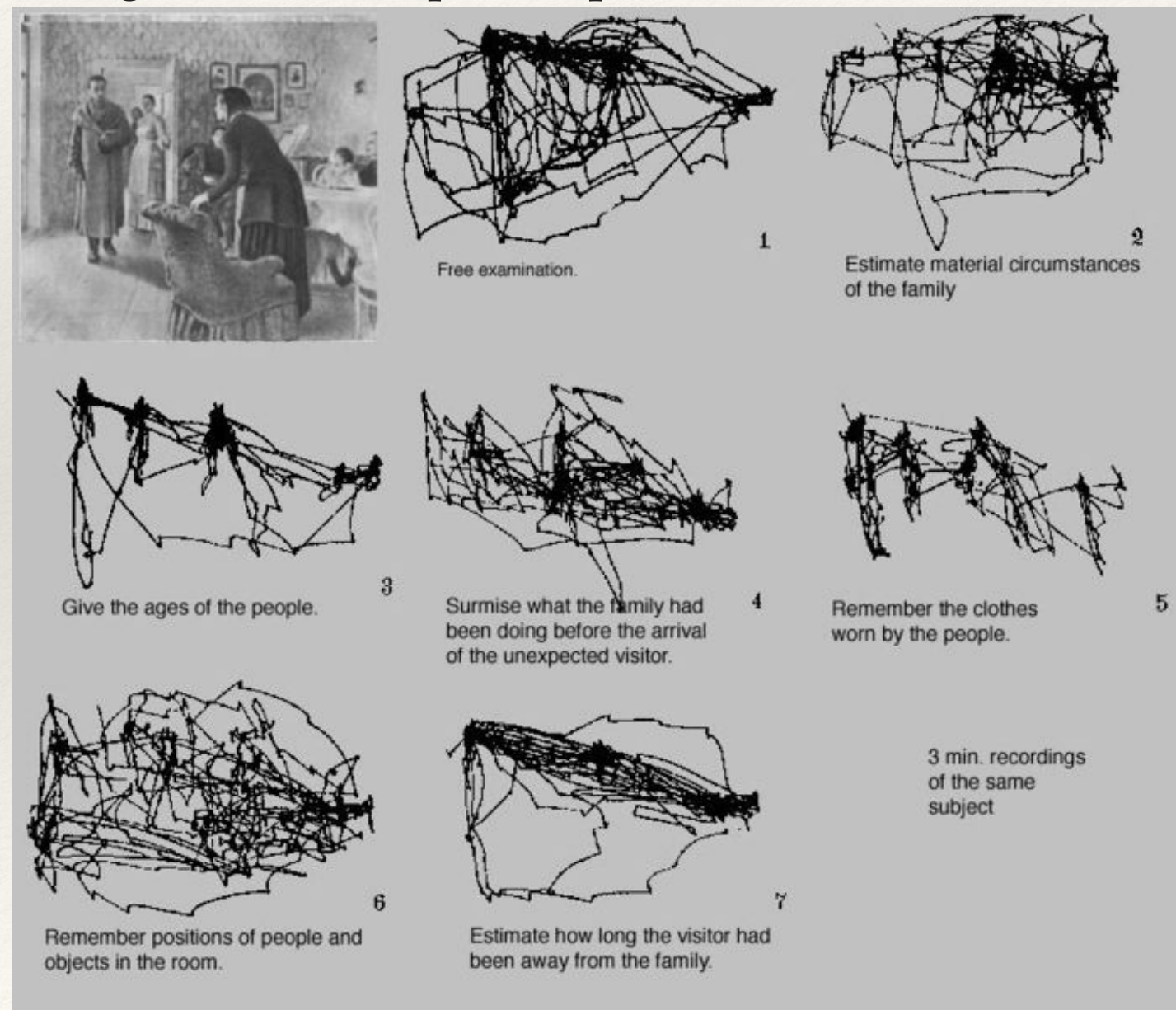
Resolving Power

- ❖ The fovea itself is a circular indentation in the retina of about 1.5mm in diameter
- ❖ By approximating fovea as a square sensor array of 1.5mm x 1.5mm because camera model assumes square or rectangular arrays of sensing elements
 - ❖ The density of cones in that area is $\sim 150,000$ elements per mm^2
- ❖ Based on these approximations, the number of cones in the region of highest acuity in the eye is about 337,000 elements
- ❖ In terms of raw resolving power, a charge-coupled device (CCD) imaging chip of medium resolution can have this number of elements in a receptor array no larger than 5mm x 5mm



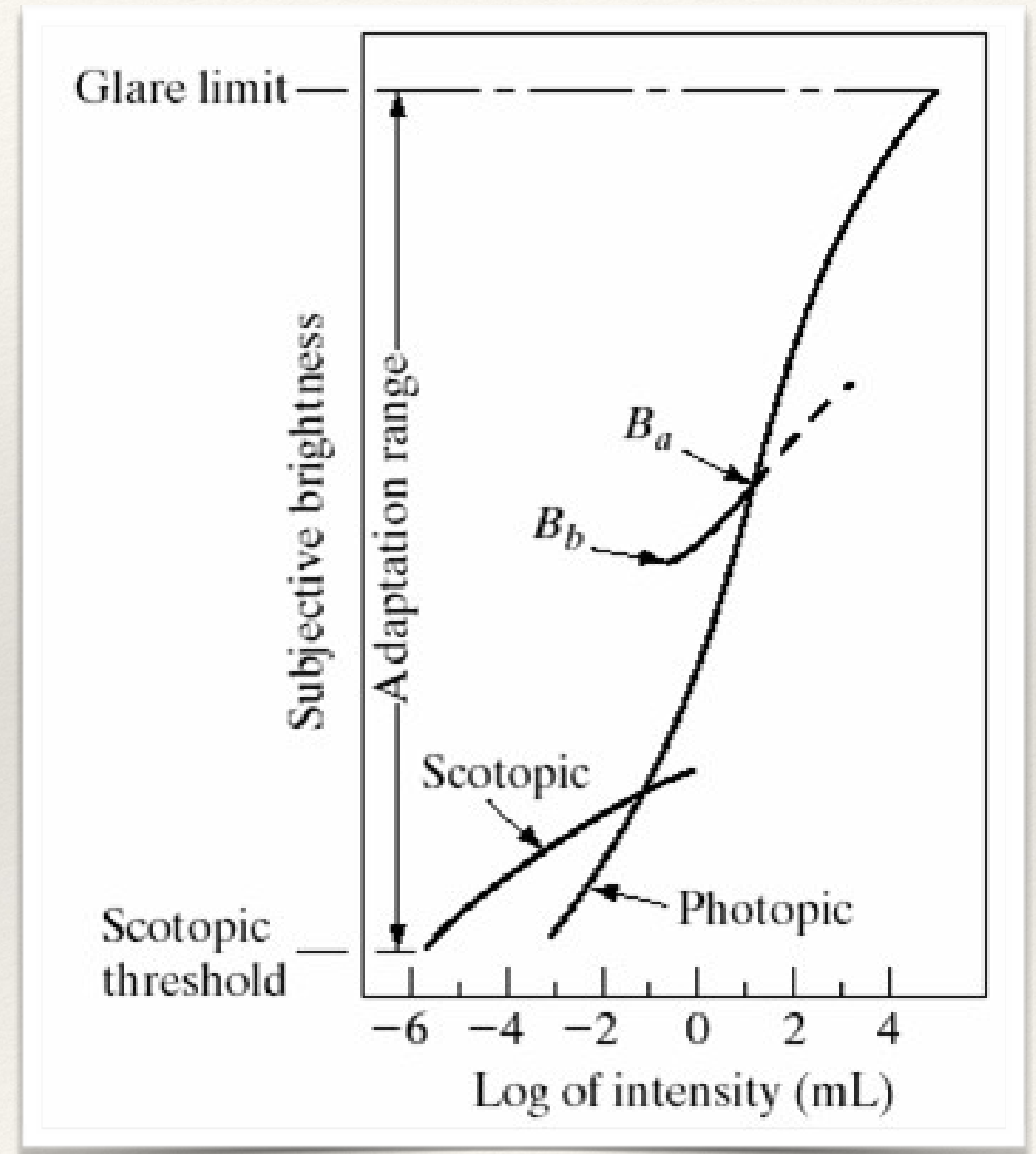
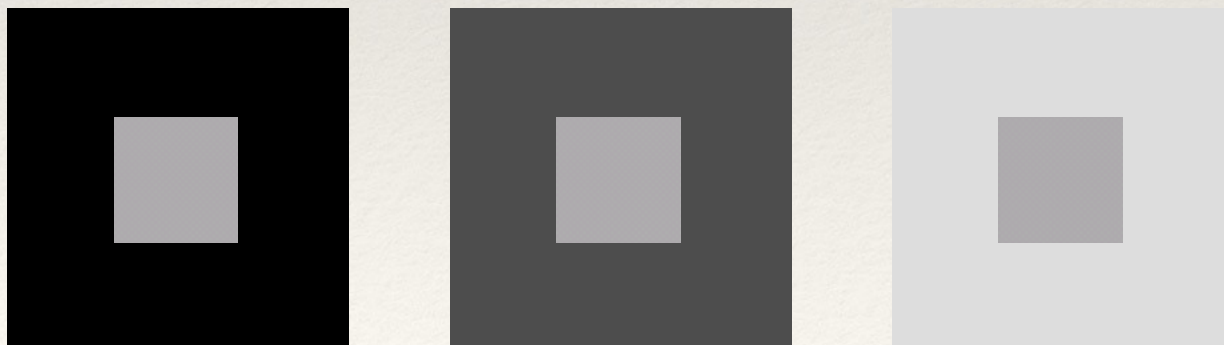
Eye Movement in Scene Viewing

- ❖ Our eyes are never stationary when interpreting the scene, our brain reconstruct a higher resolution image from multiple snapshots



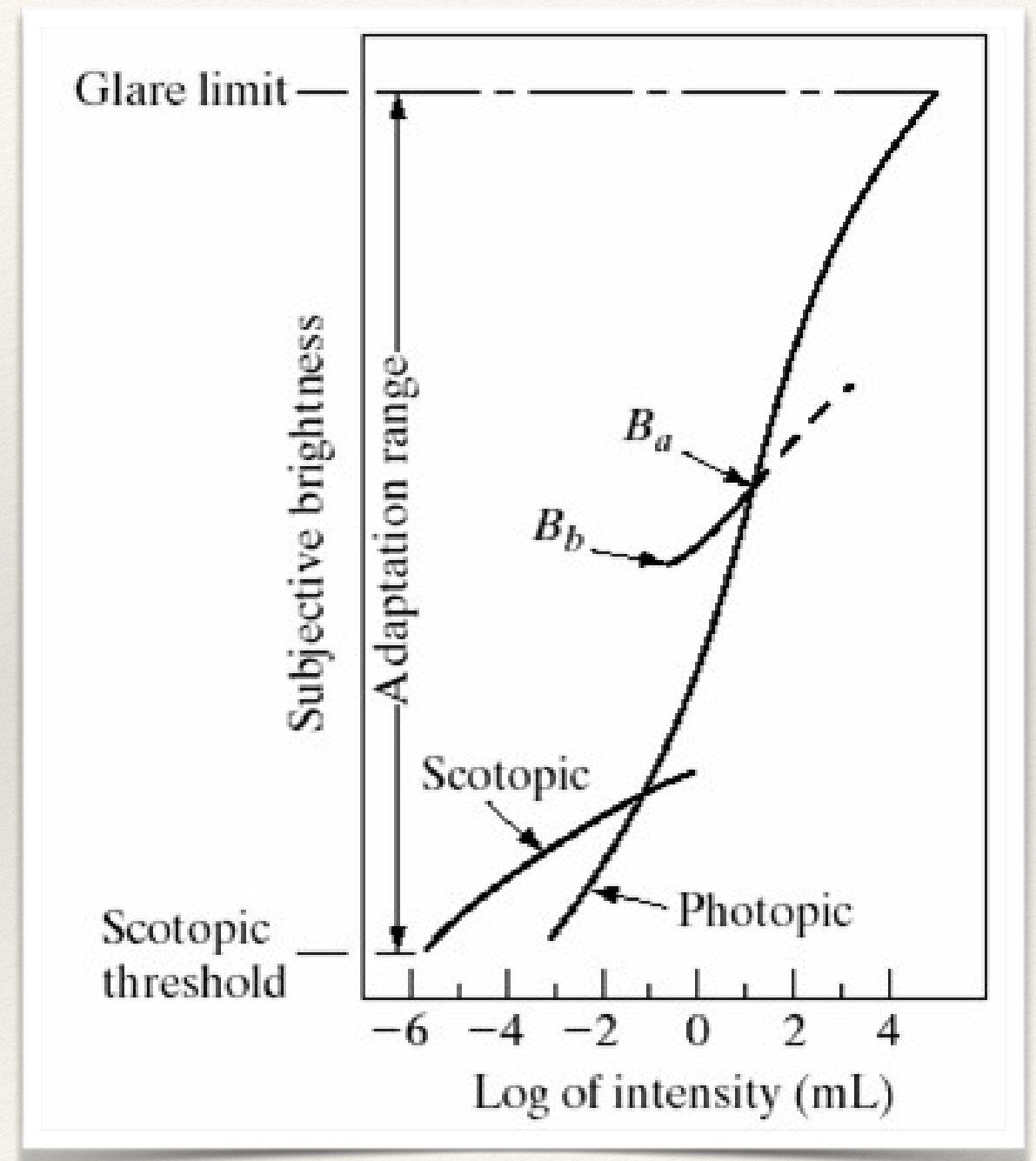
Perception of Brightness

- ❖ Subjective brightness
 - ❖ intensity as perceived by the human vision system
 - ❖ a logarithmic function of the light intensity incident on the eye
- ❖ Perceived brightness is *NOT* a simple function of intensity (see examples)



Brightness Adaptation

- ❖ Human vision system
 - ❖ can adapt an enormous range of light intensity (order of 10^{10} !)
 - ❖ cannot operate over the full range simultaneously
 - ❖ accomplishes such an impressive range by changing its overall sensitivity, a phenomenon known as *brightness adaptation*

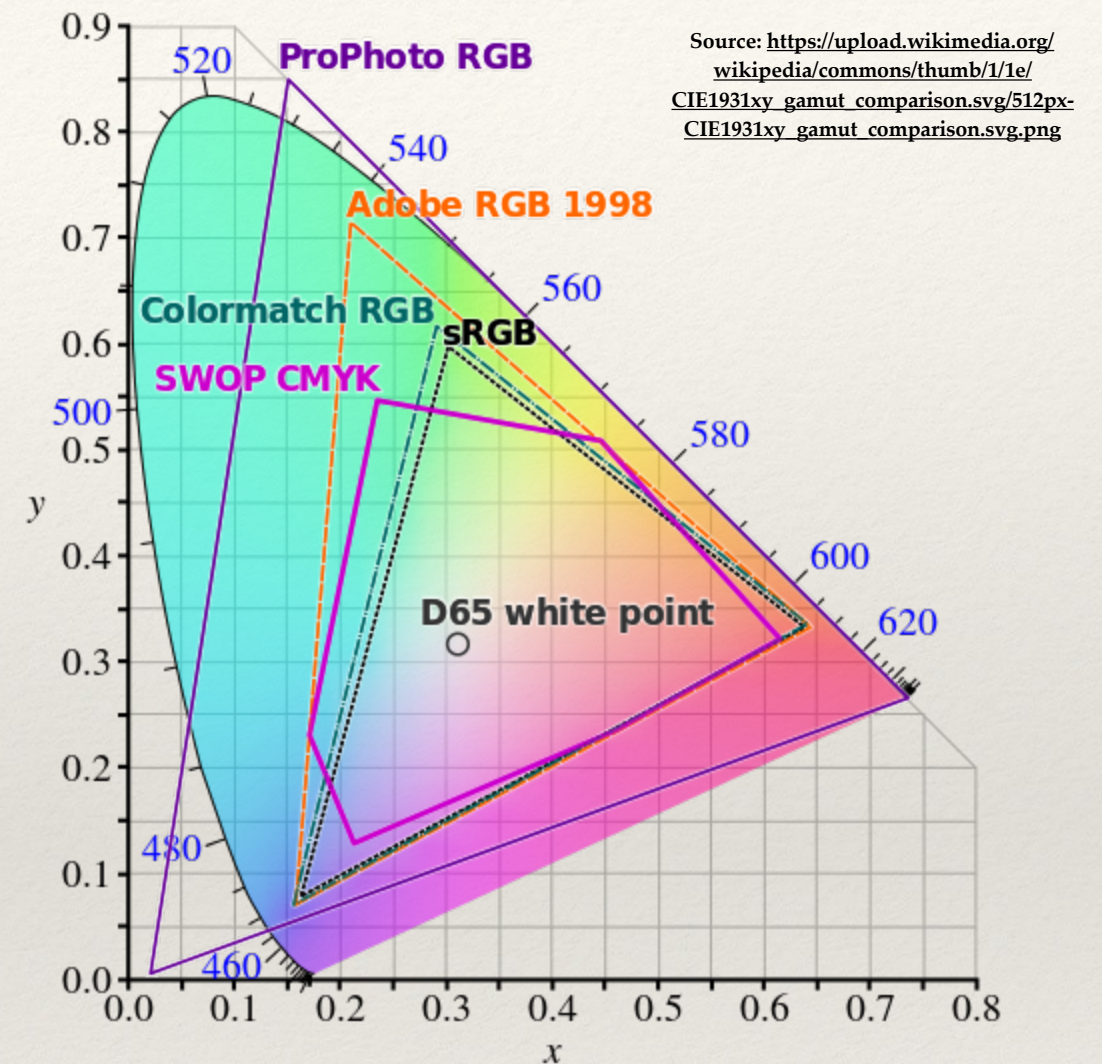


Color Spaces

- ❖ For printing industry, CMYK color space is preferred
- ❖ For display, RGB(A) color space is widely used
- ❖ For image compression, YUV color space has been predominant

$$\begin{bmatrix} Y' \\ U \\ V \end{bmatrix} = \begin{bmatrix} 0.299 & 0.587 & 0.114 \\ -0.14713 & -0.28886 & 0.436 \\ 0.615 & -0.51499 & -0.10001 \end{bmatrix} \begin{bmatrix} R \\ G \\ B \end{bmatrix}$$

$$\begin{bmatrix} R \\ G \\ B \end{bmatrix} = \begin{bmatrix} 1 & 0 & 1.13983 \\ 1 & -0.39465 & -0.58060 \\ 1 & 2.03211 & 0 \end{bmatrix} \begin{bmatrix} Y' \\ U \\ V \end{bmatrix}$$



Image

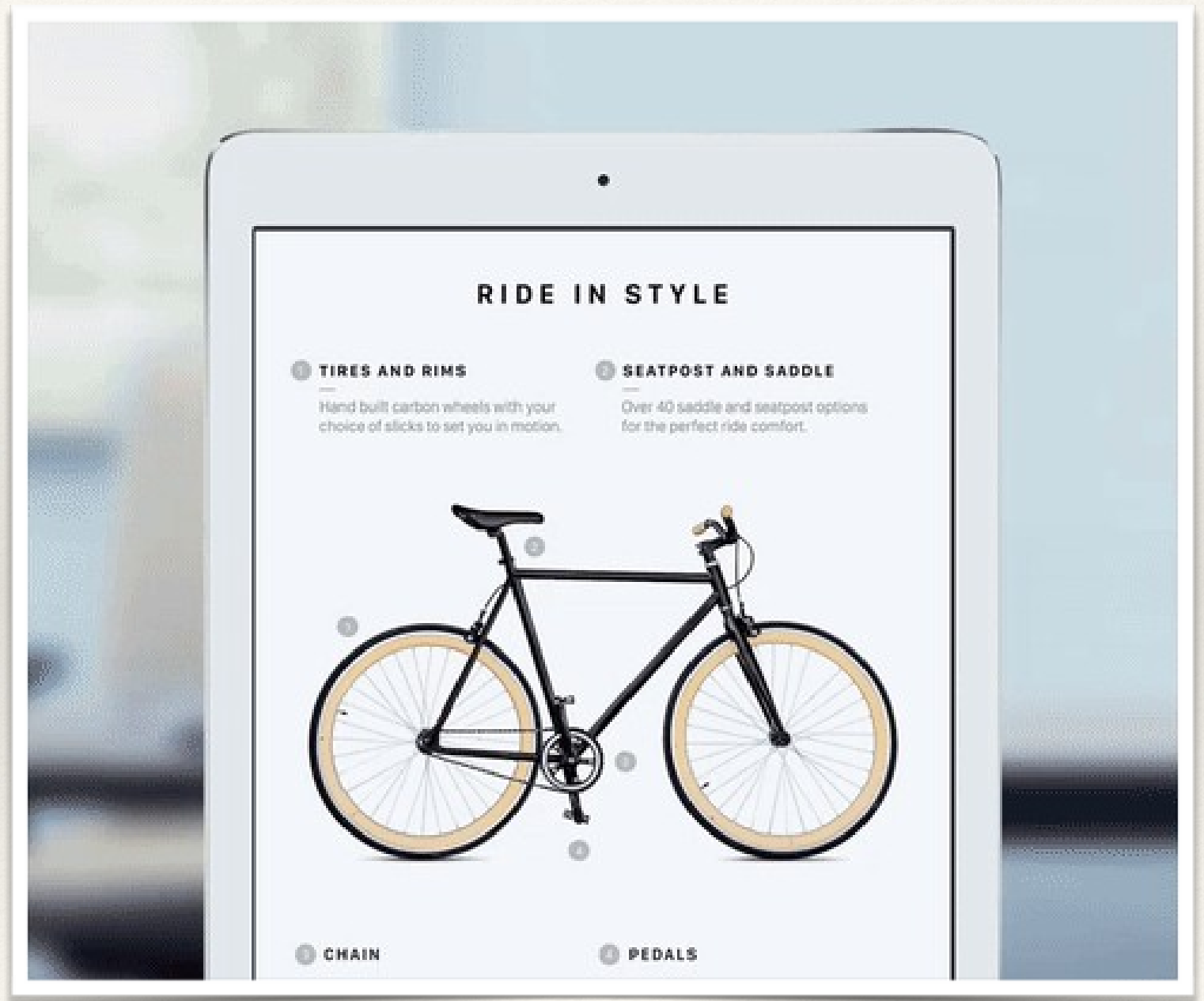
Y-Component

U-Component

V-Component

True Tone Display

- ❖ The idea behind the whole concept is that whites tend to look different under different light, but with True Tone enabled, the display can shift how the display looks no matter the lighting.

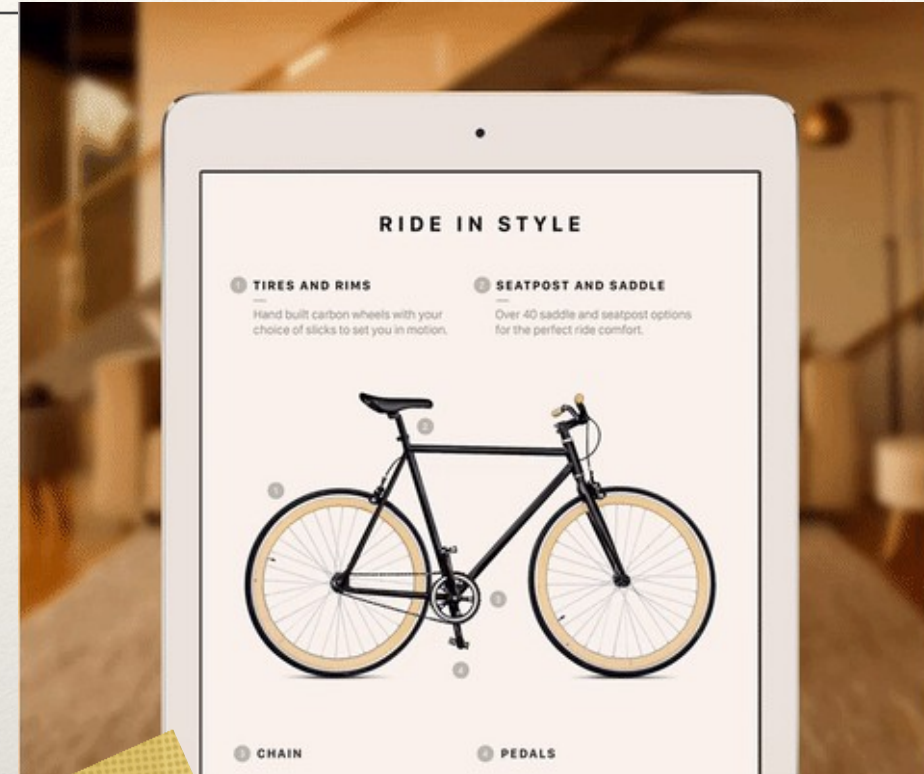
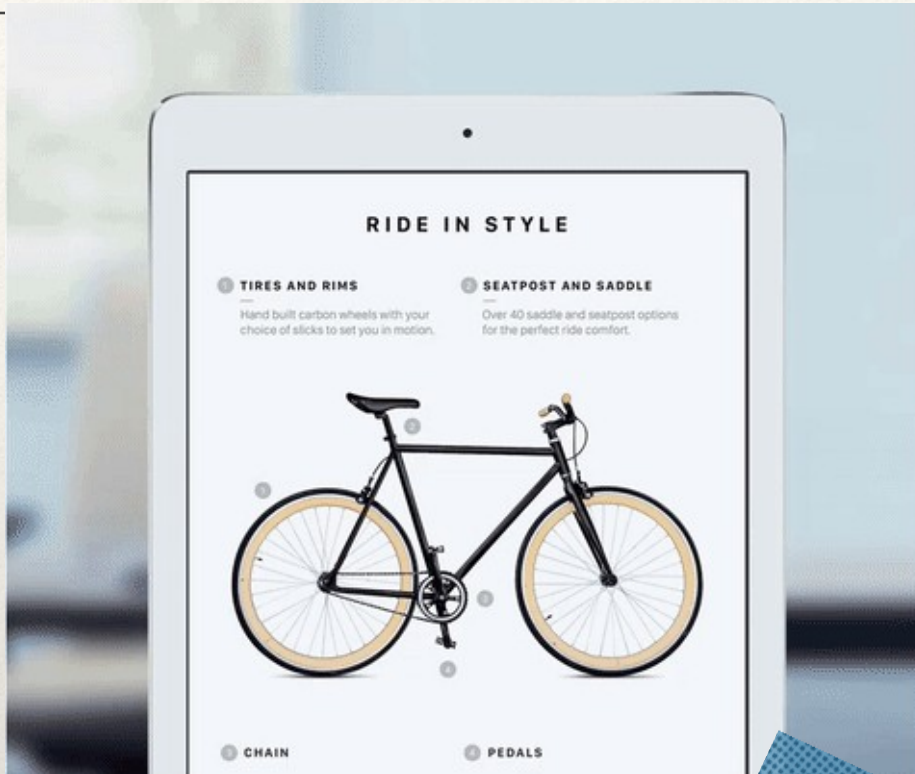


Source: <http://www.popsci.com/how-apple-crafted-ipad-pros-true-tone-display>



True Tone Display (Cont'd)

With
True Tone



Without
True Tone

